



Ground Water Department,
Rajasthan

Hydrogeological Atlas of Rajasthan Pratapgarh District



European Union
State Partnership Programme



2013



Rolta India Limited

Message



Shri Ashok Gehlot,
Hon'ble Chief Minister, Rajasthan

Ground water is an inseparable aspect in the lives of the people of Rajasthan. There are no major or perennial rivers flowing through the State. Moreover, most part of the state receives very less rainfall. Our dependence on ground water is extremely critical not only as drinking water for human beings and cattle but also as major source of irrigation.

The Government of Rajasthan is keen on sustainable utilization of the ground water. The State water policy, adopted in 2010 emphasizes on community participation in ground water management and use of information technology for widespread dissemination of data for the benefit of all sections of the society.

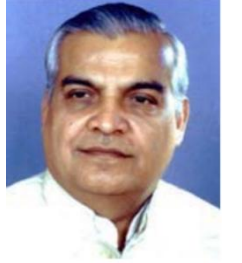
I am pleased to note that the Ground Water Department of the State has carried a detailed assessment of quality, quantity and distribution of ground water resources of the State through mapping of aquifers and using most modern tools and techniques.

It is also heartening to note that the outcome of the project is being published in the form of 'Hydrogeological Atlas of Rajasthan'. It will incorporate the scientific findings and reliable information in various types of readily understandable GIS maps based on ground water potential viz. zone maps, depth to water level maps, ground water categorization maps, chemical quality maps, aquifer maps etc. These would be certainly found useful by experts and user agencies. The same would also help in raising the awareness levels of the various stakeholders in utilizing the precious ground water resources of the State.

I take this opportunity to congratulate the Ground Water Department officials for bringing this useful publication and wish them success in all such endeavor.


(Ashok Gehlot)

Message

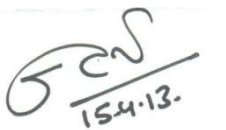


Dr. Jitendra Singh,
Hon'ble Minister for Energy,
Non-Conventional Energy Resources,
PHED, GWD, Information & Public Relations

Ground water is a renewable resource but its long term sustainability depends on the balance of recharge and withdrawal ground water is added every year through rainfall and other sources. In Rajasthan, volume of water extracted for different purposes far exceeds its replenishment which has led to depletion of ground water in large parts of state.

While the surface water is visible and largely measurable, the ground water is hidden underneath. To meet the ever increasing demand of water, the state has been promoting conjunctive use of water. In order to have a state level estimation of water resources, the estimation of ground water quality and quantity is a key input. The studies being very scientific in nature it remains a challenge how to communicate the message to all stakeholders.

The publication of '*Hydrogeological Atlas of Rajasthan*' is a major step forward and will help in visualization of ground water related information in very simplified graphic and tabular forms. The effort that has gone into creation of the atlas from basic information in archives is very much appreciated.


15.4.13.

(Jitendra Singh)

Message



Dr. Purushottam Agarwal

Principal Secretary to Government,

Public Health Engineering and Ground Water Department, Rajasthan

Ground water has been the principal source over years to meet the needs of human and cattle population of the state. Increasingly modern techniques are used for extraction of large volume of ground water which has exceeded recharge in most parts of the State. As a result, water table in most of the areas has gone down significantly. People are digging deeper and deeper to reach the ground water to meet their increasing needs. Maintaining a healthy practice of balance between recharge and withdrawal is critical for its sustainability.

The state had initiated various programs under which a systematic assessment of water resources is being carried out and all future water management plans are chalked out on the basis of sound database and historic data analysis. Increasing population, industrialization, agricultural need, power generation etc. are bound to pose challenges before us in coming years and to face them we have documented 'Water Resource Vision 2045' which highlights the short term (upto 2015) and long term (upto 2045) thrust areas and action plan which are pre-requisites for successful implementation of the State Water Policy. We are committed to achieving the objective of optimum use of every drop of scarce and precious utilizable water resource. In this regard, the Ground Water Department has also carried out a systematic assessment of ground water resources and three dimensional mapping of aquifers that gives a new insight.

I congratulate the Ground Water Department on publication of 'Hydrogeological Atlas of Rajasthan' which emphasizes on various aspects of ground water which I feel will help us in spreading the information on current scenario and consequently help in managing it better.

(Purushottam Agarwal)

Foreword



Arno Schaefer

Minister Counselor, Head of Operations

The Government of Rajasthan has entered into an agreement with the European Union through the State Partnership Programme for implementing state-wide water sector reform leading to sustainable integrated water resources management in Rajasthan, and supporting the Panchayat Raj institutions in executing their responsibilities in equitable access to safe, adequate, affordable and sustainable water supply as well as conservation, stabilization and replenishment of surface and ground water.

Ground water is the primary source for drinking, domestic, agriculture and industries and other purposes in Rajasthan. The stage of ground water development is close to 135% i.e., if the net annual ground availability in Rajasthan is about 10.7 billion cubic meter, the annual draft is 14.5 billion cubic meter. As a result of this, the water levels have been depleting alarmingly. The ground water in 166 out of 249 blocks of the state fall under the category of over-exploited. This indiscriminate and sometime exploitative use of ground water has led to questions regarding its sustainability.

This publication '*Hydrogeological Atlas of Rajasthan*' is the culmination of a long term effort of collation, integration, interpretation and analysis of vast archive of historic data obtained by Rajasthan Ground Water Department. The atlas is in three parts, (1) Statewide Atlas, which shows the state of ground water, water quality, current stage of exploitation, aquifer mapping and basic information on administrative setup, climate, geology and geomorphology; (2) A detailed series, wherein compilation of information and district level thematic maps and all 33 districts have been separately presented and (3) Compilation of information at the River Basin level. In addition to the thematic maps, a large number of cross sections and the 3 dimensional features of aquifers have also been presented.

I am convinced that the Atlas will contribute to better understand ground water and help managerial and future planning. The current Geographic Information System (GIS) based study is a pioneering step in India and will pave the way for similar approach to be adopted by other Indian States.

I appreciate very much the efforts made by the Ground Water Department, Government of Rajasthan, The European Union Technical Assistance Team and M/s Rolta India Limited for undertaking this study and compiling this Atlas under the India-EU State partnership Programme.

(Arno Schaefer)

Hydrogeological Atlas of Rajasthan

Pratapgarh District

Contents:



List of Plates	Title	Page No.
Plate I	Administrative Map	2
Plate II	Topography	4
Plate III	Rainfall Distribution	4
Plate IV	Geological Map	6
Plate V	Geomorphological Map	6
Plate VI	Aquifer Map	8
Plate VII	Stage of Ground Water Development (Block wise) 2011	8
Plate VIII	Location of Exploratory and Ground Water Monitoring Stations	10
Plate IX	Depth to Water Level (Pre-Monsoon 2010)	10
Plate X	Water Table Elevation (Pre-Monsoon 2010)	12
Plate XI	Water Level Fluctuation (Pre-Post Monsoon 2010)	12
Plate XII	Electrical Conductivity Distribution (Average Pre-Monsoon 2005-09)	14
Plate XIII	Chloride Distribution (Average Pre-Monsoon 2005-09)	14
Plate XIV	Fluoride Distribution (Average Pre-Monsoon 2005-09)	16
Plate XV	Nitrate Distribution (Average Pre-Monsoon 2005-09)	16
Plate XVI	Depth to Bedrock	18
Plate XVII	Map of Unconfined Aquifer	18
Glossary of terms		19

2013

ADMINISTRATIVE SETUP

DISTRICT – PRATAPGARH

Location:

Pratapgarh district is located in the southern part of Rajasthan. It is bounded in the northwest by Chittaurgarh district, in the west by Udaipur and Dungarpur districts, southwest by Banswara district, in the east and southeast by state of Madhya Pradesh. It stretches between 23° 31' 49.84" to 24° 30' 16.57" north latitude and 74° 13' 19.93" to 74° 58' 59.58" east longitude covering area of 4,400.7 sq km. The district is systematically drained by three prominent rivers and major part of the district in the centre and west is part of 'Mahi River Basin' whereas the north-south extending strip in the eastern border is part of 'Chambal River Basin' and the small part in the north is part of 'Banas River Basin'. Most of the district is hilly/rocky of which about 34% is prominently hilly.

Administrative Set-up:

Pratapgarh district is administratively divided into five blocks. The following table summarizes the basic statistics of the district at block level.

S. No.	Block Name	Population (Based on 2001 census)	Area (sq km)	% of District Area	Total Number of Towns and Villages
1	Arnod	1,19,837	694.5	15.8	179
2	Chhoti Sadri	1,16,676	704.8	16.0	152
3	Dhariawad	1,52,615	891.6	20.3	162
4	Peepalkhoont	80,988	545.4	12.4	94
5	Pratapgarh	2,36,651	1,564.3	35.5	359
Total		7,06,767	4,400.7	100.0	946

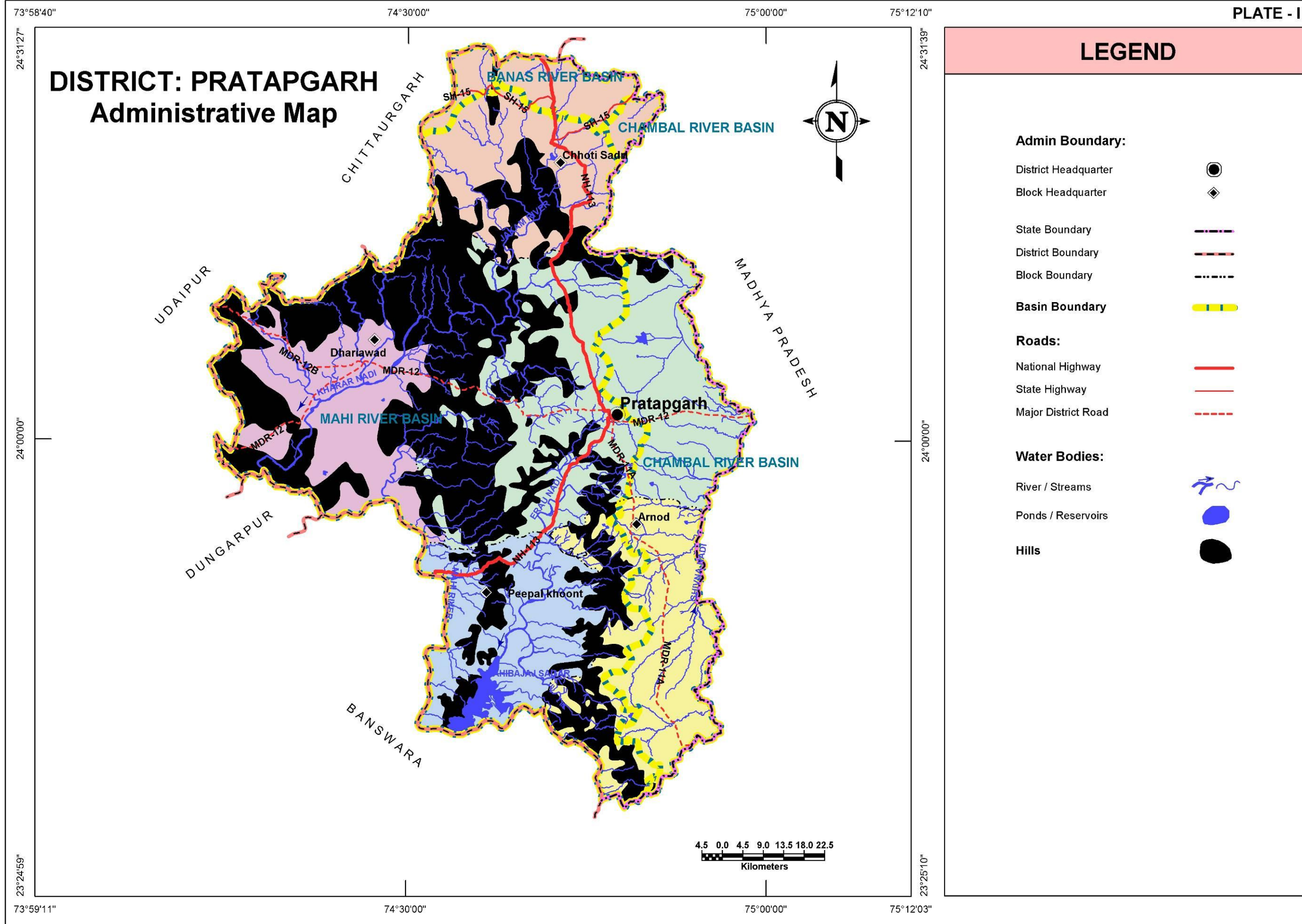
Pratapgarh district has 946 towns and villages, of which five are block headquarters as well.

Climate:

The climate of Pratapgarh district is more similar to Madhya Pradesh than to typically arid Rajasthan. The district witnesses three distinct seasons viz, winters, summer and monsoon. In winter season, extending between November to February, average temperature is 18 °C whereas in summer months of March to June, the average temperature is 31 °C. Summers are dry and hot with temperatures soaring to 45 °C in the months of June. Monsoon months between July and Sept-Oct are wet and humid as the district receives fairly good rainfall. The average total annual rainfall is 896.2 mm.



PLATE - I



TOPOGRAPHY

Southwestern part of the district has some undulating plains but most of the remaining areas of the district are hilly terrain. The general slope of the terrain is from east and northeast towards southwest. However, the whole of north-south trending strip in the eastern part of the district that is part of Chambal river basin has west to east slope and the small part in the north which is part of Banas basin has northerly slope. The topographic elevation shows wide variation ranging from the lowest of 161.9 m above mean sea level in Dhariawad block in the western part of the district to the highest of 576.5 m above mean sea level in Pratapgarh in eastern part of the district.

Table: Block wise minimum and maximum elevation

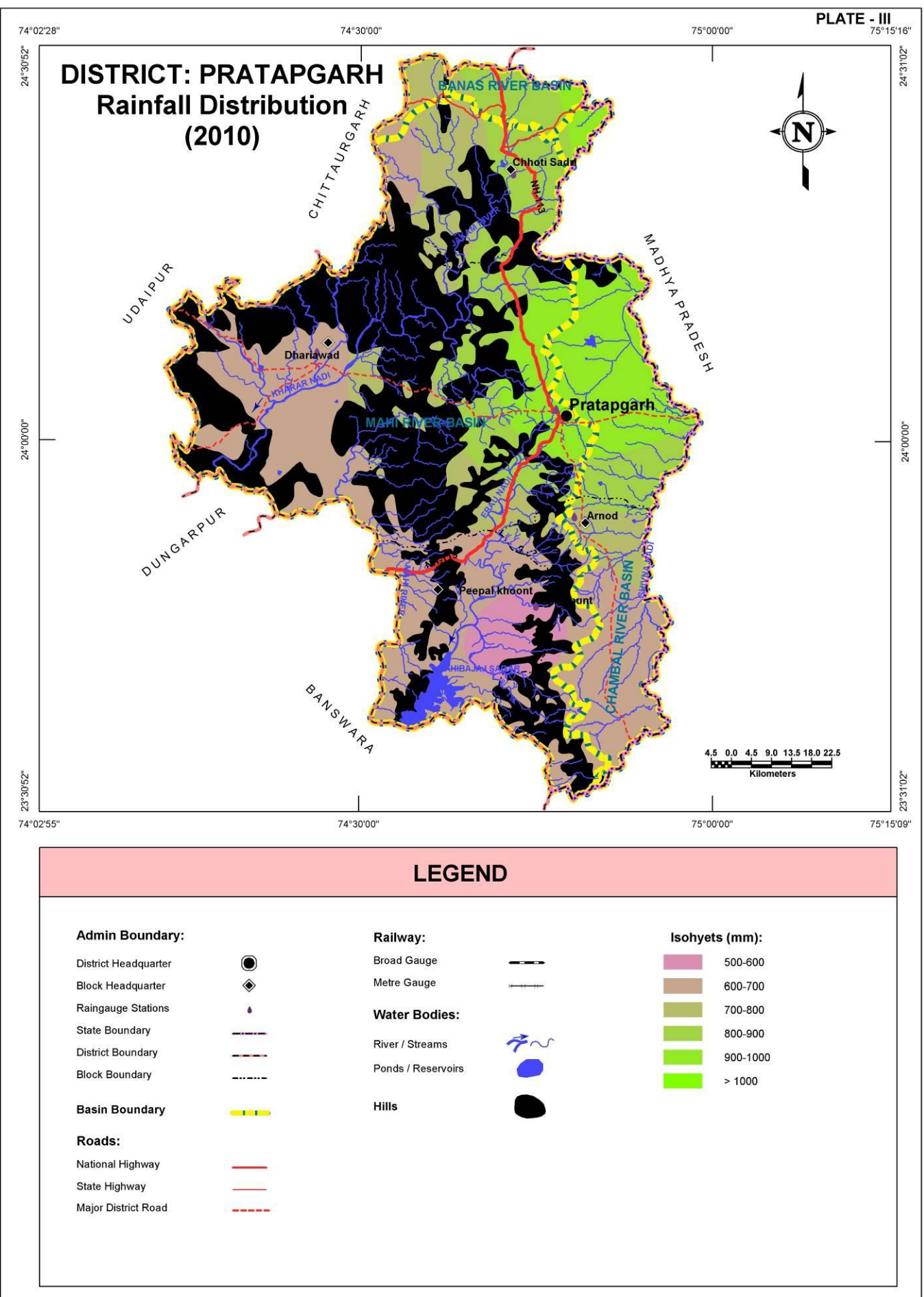
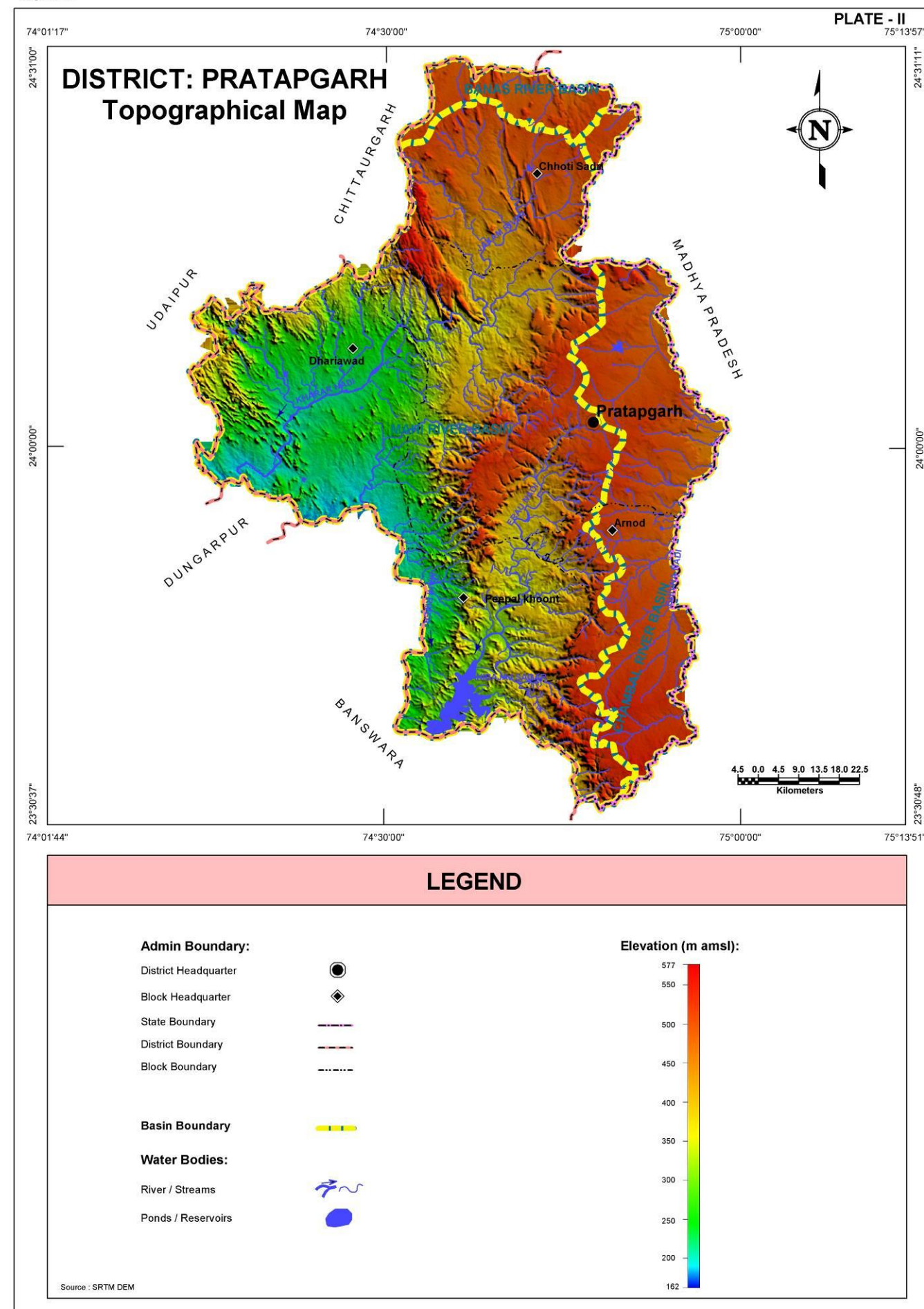
S. No.	Block Name	Min. Elevation (m amsl)	Max. Elevation (m amsl)
1	Arnod	348.4	576.4
2	Chhoti Sadri	386.8	576.1
3	Dhariawad	161.9	495.2
4	Peepalkhoont	187.4	529.8
5	Pratapgarh	204.9	576.5

RAINFALL

The district receives a good rainfall. The general distribution of rainfall across the district can be visualized from isohyets presented in the Plate – III, where it can be seen that most of the district received rainfall in the range of 600-900mm in year 2010. The annual average rainfall was 717.5 mm based on the data of available blocks. Highest annual rainfall was observed in Pratapgarh block (1002 mm) whereas lowest was in Peepalkhoont block (563.2 mm). The highest average annual rainfall noticed in Pratapgarh block (844.5 mm).

Table: Block wise annual rainfall statistics (derived from year 2010 meteorological station data)

Block Name	Minimum Annual Rainfall (mm)	Maximum Annual Rainfall (mm)	Average Annual Rainfall (mm)
Arnod	581.2	821.7	669.3
Chhoti Sadri	615.7	948.7	794.5
Dhariawad	605.8	753	660.7
Peepalkhoont	563.2	702.4	618.5
Pratapgarh	660.9	1,002.00	844.5



GEOLOGY

DISTRICT – PRATAPGARH

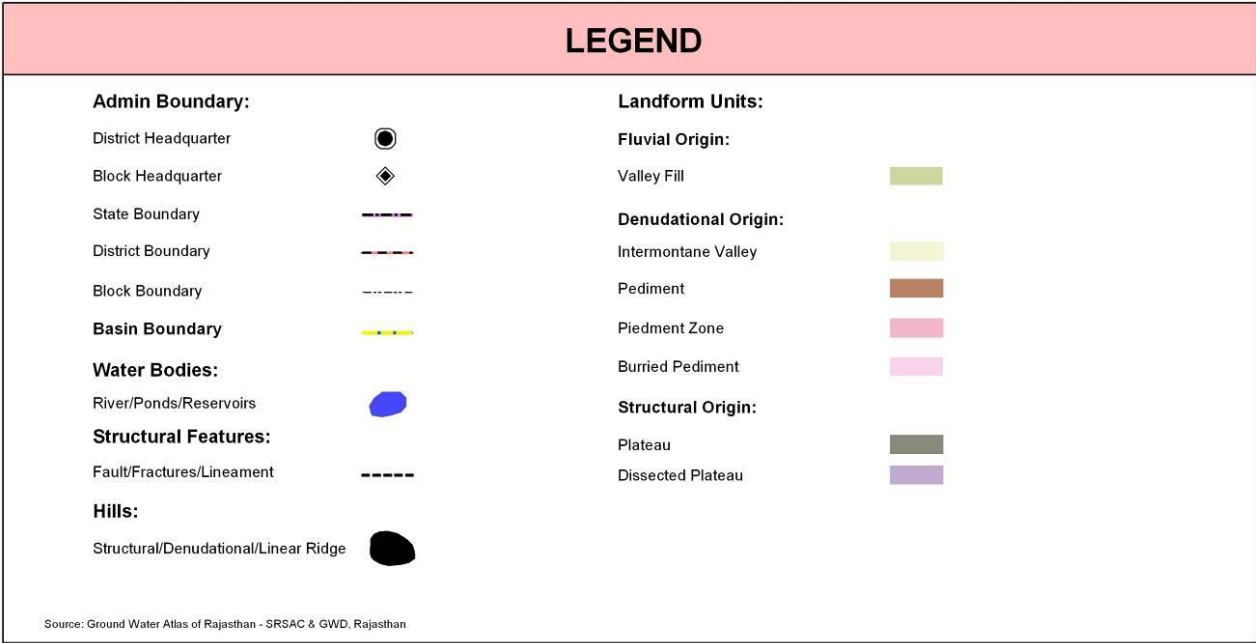
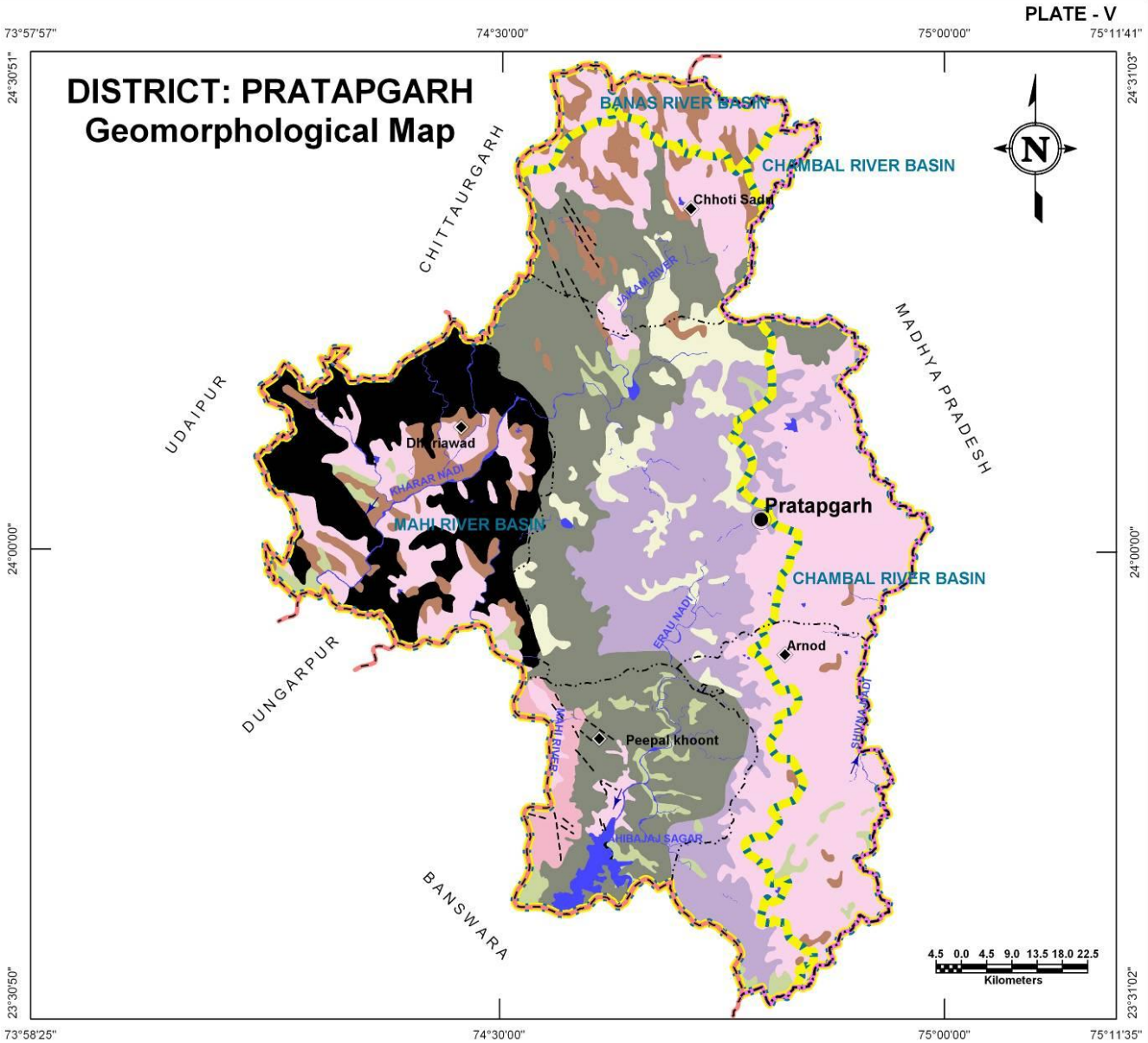
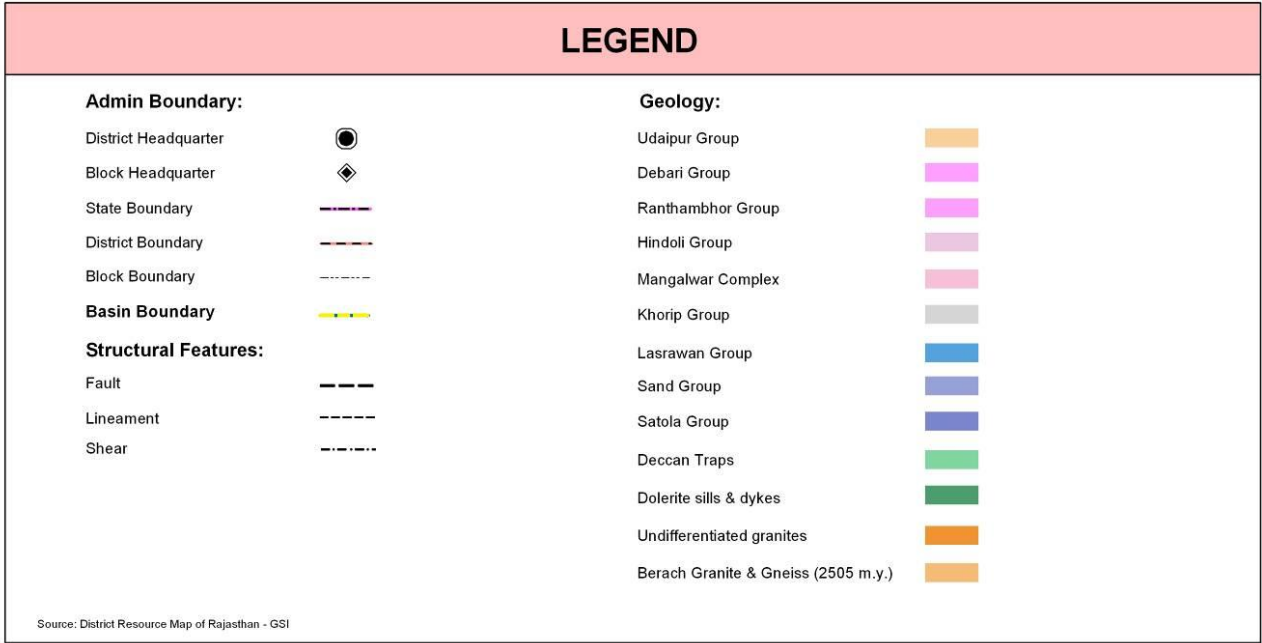
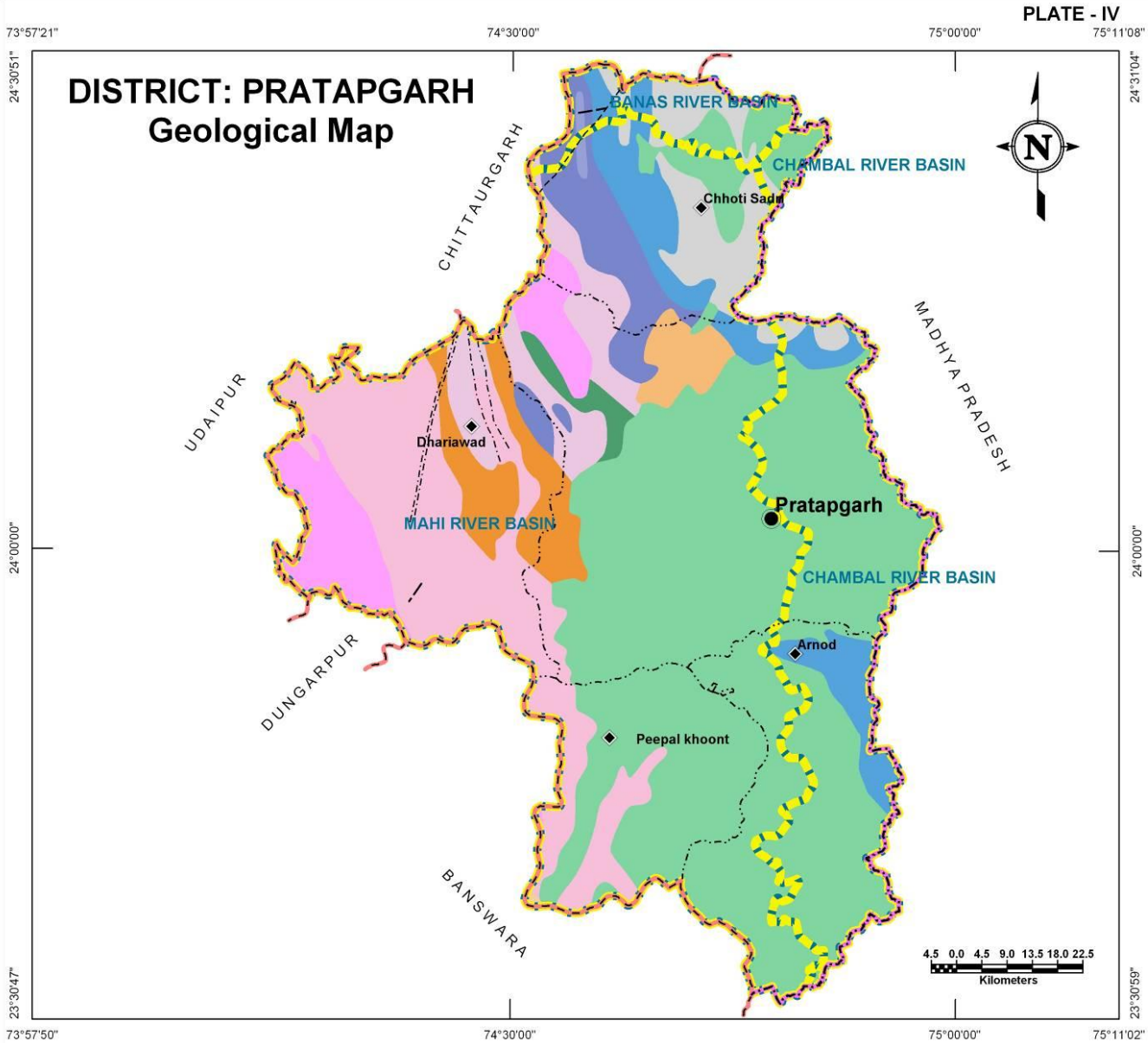
The district exposes rocks belonging to Bhilwara, Aravalli and Vindhyan Super Groups. Deccan traps occupy significant area of the district in the central, eastern and southern parts. The Bhilwara Super Group is divided into Ranthambhor, Hindoli Groups and Mangalwar complex which consist Shale, slates, phyllites, limestone, dolomitic marble, migmatites, feldspathic mica schist and amphibolites. It is exposed in northwestern, western and southwestern parts of the district. The Aravalli Super Group is divided into Udaipur and Debari Group which consists of the rock formation such as Phyllite, mica schist, quartzite, dolomite and migmatites. The Vindhyan Super Group is divided in Khorip, Lasrawan, Sand and Satola Groups with Shale, Conglomerate, Limestone, sandstone and siltstone formations. Vindhyan Super Group is exposed mainly in northern part of the district and some exposure in southeastern parts in Chhotisadri block and also partly in Dhariawad, Pratapgarh and Arnod blocks.

Age/Super Group	Group	Formation
Cretaceous		Deccan Traps
Vindhyan	Khorip	Shales, Sand-Stone, Conglomerate, Porceltanite
	Lasrawan	Limestone, Shale, Sandstone, Siltstone
	Sand	
	Satola	
--X-----X-----X-----X---Unconformity---X-----X-----X-----X--		
Aravalli	Udaipur	Phyllite, mica schists, meta siltstone, quartzite, Dolomite, gneisses and migmatites.
	Debari	Meta-arkose, quartzite, phyllite, dolomitic marble and dolomite.
--X-----X-----X-----X---Unconformity---X-----X-----X-----X--		
Bhilwara	Ranthambhor	Shale and slates
		Dolerite sills & dykes
	Hindoli	Shale, Slates, Phyllites, Metagreywackes, Limestone, Dolomitic marble
	Mangalwar Complex	Migmatites, gniess, Feldspathic mica schist, Garnet-mica schist, impure Marble and Para-amphibolite.
	Berach Granite & Gneiss	

GEOMORPHOLOGY

Table: Geomorphologic units, their description and distribution

Origin	Landform Unit	Description
Denudational	Buried Pediment	Pediment covers essentially with relatively thicker alluvial, colluvial or weathered materials.
	Intermontane Valley	Depression between mountains, generally broad & linear, filled with colluvial deposits.
	Pediment	Broad gently sloping rock flooring, erosional surface of low relief between hill and plain, comprised of varied lithology, criss-crossed by fractures and faults.
	Pediment Zone	Formed by coalescence of several alluvial fans by stream covering large area at foot hills, with gentle slope in humid to sub humid region.
Fluvial	Valley Fill	Formed by fluvial activity, usually at lower topographic locations, comprising of boulders, cobbles, pebbles, gravels, sand, silt and clay. The unit has consolidated sediment deposits.
Structural	Dissected Plateau	Plateau, criss-crossed by fractures forming deep valleys.
	Plateau	Formed over varying lithology with extensive, flat, landscapes, bordered by escarpment on all sides. Essentially formed horizontally layered rocky marked by extensive flat top and steep slopes. It may be criss crossed by lineament.
Hills	Denudational, Structural Hill, Linear Ridge	Steep sided, relict hills undergone denudation, comprising of varying lithology with joints, fractures and lineaments. Linear to arcuate hills showing definite trend-lines with varying lithology associated with folding, faulting etc. Long narrow low-lying ridge usually barren, having high run off may form over varying lithology with controlled strike.



AQUIFERS

DISTRICT – PRATAPGARH

Though the district is predominantly hilly with hills occupying about 34% of the district area, aquifers have formed in weathered, fractured and jointed hardrocks which contain good quality and quantity of water. Basalts occupy about 41% of area and occupy areas in the central, eastern and southeastern part of the district. Next most widespread aquifer is formed in BGC with 15.3% areal coverage. Shales and schist also form minor aquifers in northern part of the district.

Table: aquifer potential zones their area and their description

Aquifer in Potential Zone	Area (sq km)	% age of district	Description of the unit/Occurrence
Schist	109.7	2.5	Medium to fine grained compact rock. The litho units are soft, friable and have closely spaced cleavage.
Shale	315.7	7.2	Grey, light green and purple in colour and mostly splintery in nature.
Basalt	1824.3	41.4	Dark grey, olive green and green colour, compact, vesicular, amygdaloidal and weathered.
BGC	674.4	15.3	Grey to dark coloured, medium to coarse grained rocks.
Hills	1477.1	33.6	
Total	4,401.2	100.0	

STAGE OF GROUND WATER DEVELOPMENT

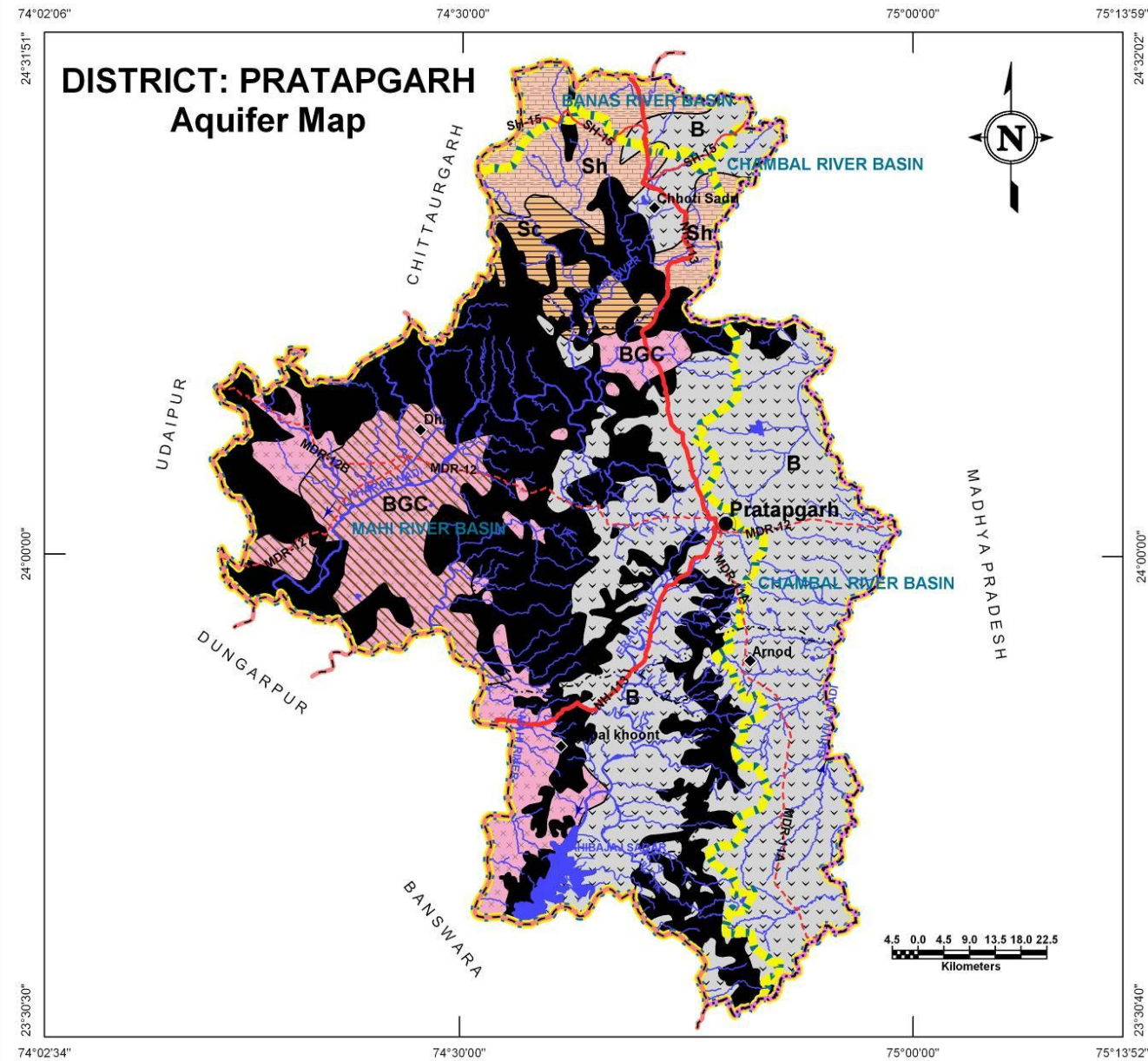
Ground water resource assessment studies have revealed that in all the blocks of the Pratapgarh district as significant development of ground water has taken place. Peepalkhoont is the only block where ground water can still be developed as it falls in 'Safe' category. All the remaining blocks are either in 'Over Exploited' or in 'Critical' category implying that the ground water resources have nearly been exhausted and no more development be resorted to in those blocks to prevent further depletion.

Categorization on the basis of stage of development of ground water	Block Name
Safe	Peepalkhoont
Critical	Dhariawad
Over Exploited	Arnod, Pratapgarh, Chhoti Sadri

Basis for categorization: Ground water development $\leq 70\%$ - Safe, $\leq 100\%$ - Critical and $>100\%$ - Over-Exploited.



PLATE - VI



LEGEND

Admin Boundary:

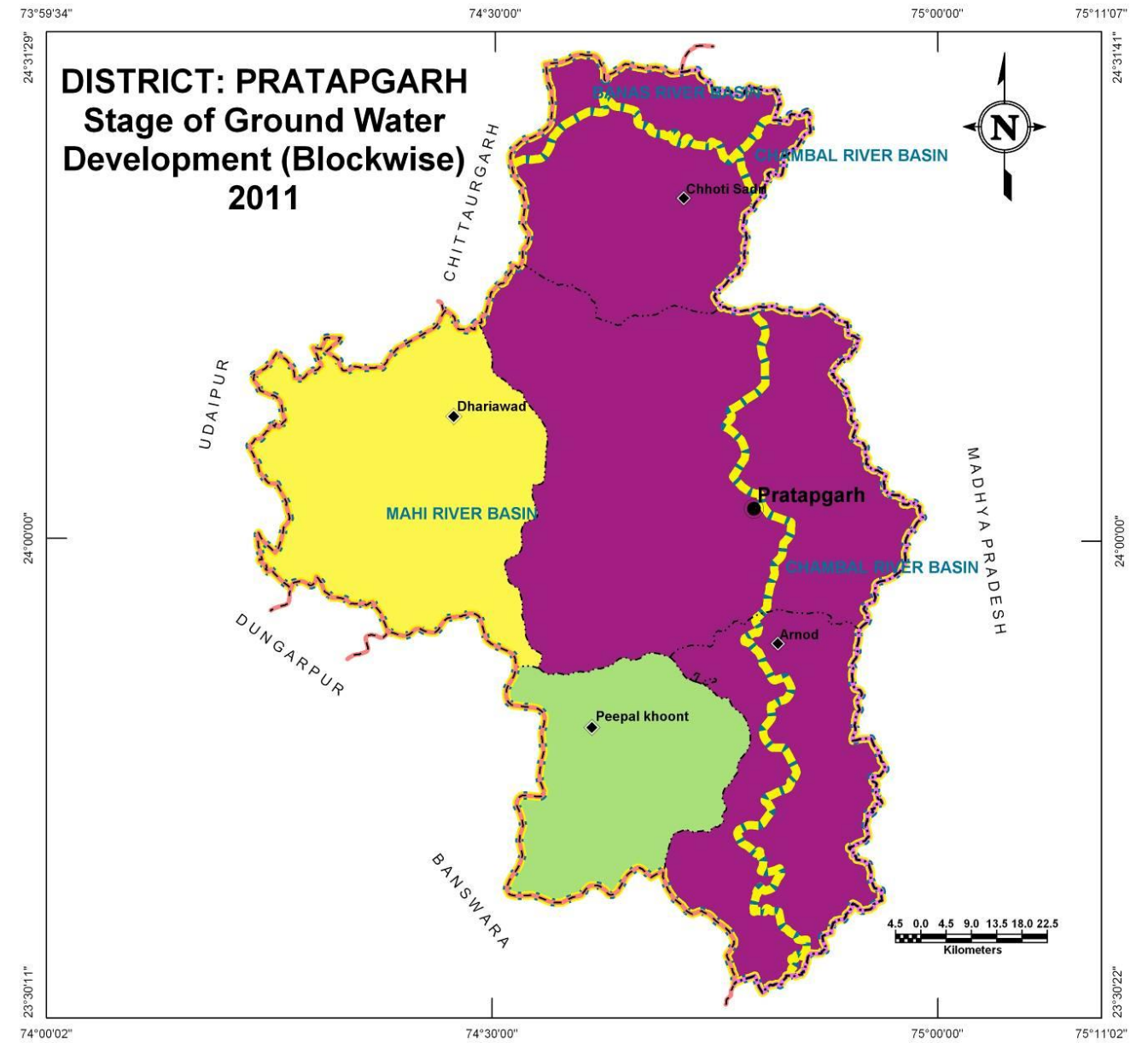
- District Headquarter
- Block Headquarter
- State Boundary
- District Boundary
- Block Boundary
- Basin Boundary
- Roads:
 - National Highway
 - State Highway
 - Major District Road

Water Bodies:

- River / Streams
- Ponds / Reservoirs
- Hills
- Command Area
- Aquifer:
 - Schist
 - Shale
 - Basalt
 - BGC

Source: Ground Water Potential Zone Map - GWD, Rajasthan

PLATE - VII



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- State Boundary
- District Boundary
- Block Boundary
- Basin Boundary

Stage of Ground Water Development:

- Safe
- Critical
- Over-Exploited

Source: Ground Water Department, Rajasthan

LOCATION OF EXPLORATORY AND GROUND WATER MONITORING WELLS

DISTRICT – PRATAPGARH

Pratapgarh district has a fairly well distributed network of exploratory wells (23) and ground water monitoring stations (123) in the district owned by RGWD (21 and 107 respectively) and CGWB (2 and 23 respectively). The exploratory wells have formed the basis for delineation of subsurface aquifer distribution scenario in three dimensions. Benchmarking and optimization studies suggest that both the ground water level and quality monitoring network need to be strengthened by adding 55 additional wells to water level monitoring network and just 7 wells to water quality network for optimization of the network.

Table: Block wise count of wells (existing and recommended)

Block Name	Exploratory Wells			Ground Water Monitoring Stations			Recommended additional wells for optimization of monitoring network	
	CGWB	RGWD	Total	CGWB	RGWD	Total	Water Level	Water Quality
Arnod	-	2	2	3	22	25	22	0
Chhoti Sadri	-	3	3	1	26	27	0	7
Dhariawad	1	3	4	4	18	22	0	0
Peepalkhoont	1	10	11	3	15	18	5	0
Pratapgarh	-	3	3	5	26	31	28	0
Total	2	21	23	16	107	123	55	7

DEPTH TO WATER LEVEL (PRE MONSOON – 2010)

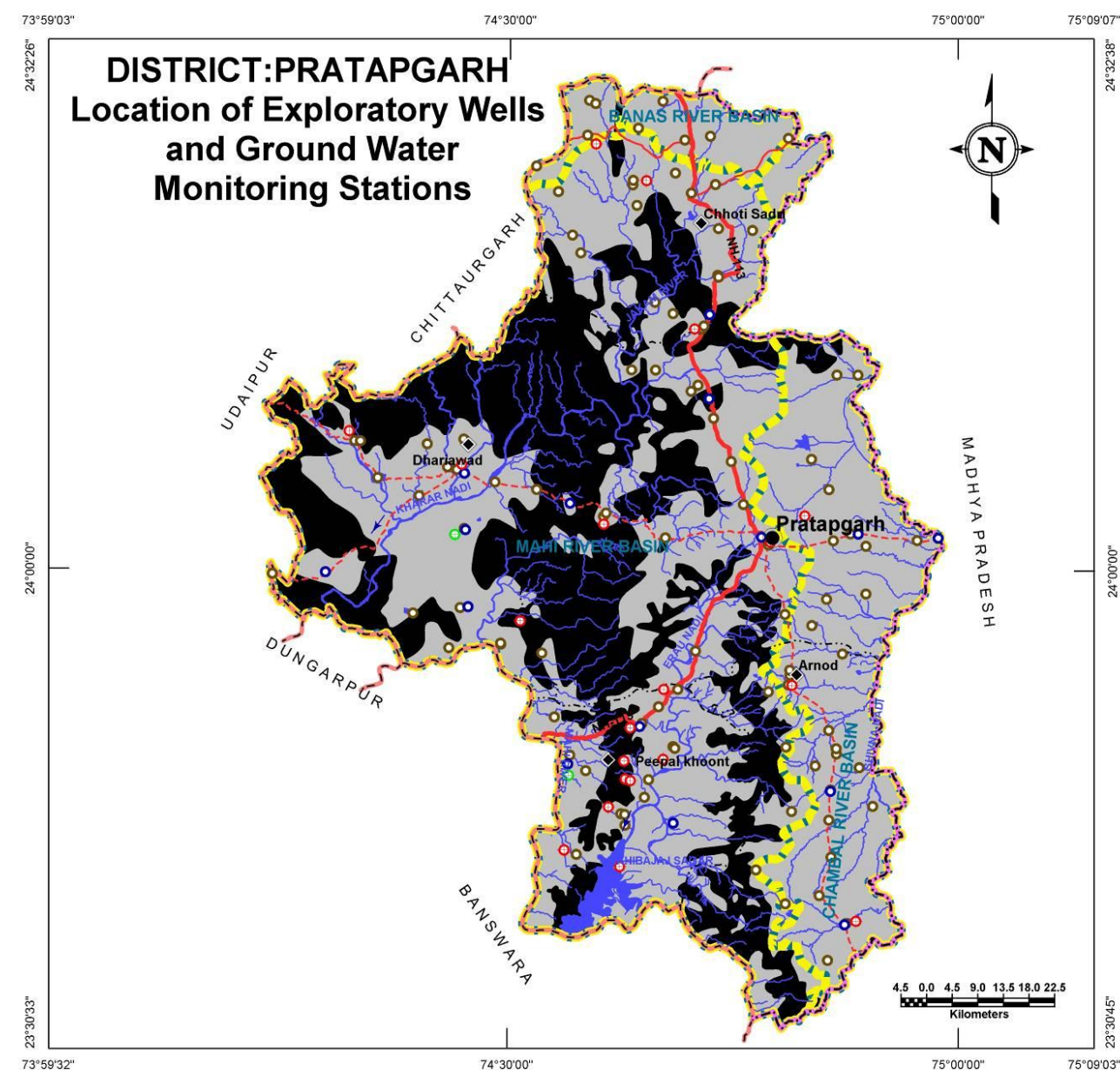
In spite of being a predominantly hard rock area, the district shows very limited variation in depth to ground water levels i.e., from less than 10m bgl to around 30m bgl. Western sides of the district are marked with moderately deeper water level in between 10-20m bgl whereas the eastern parts of the districts observed with water level more than 20m bgl. A small patch of quite shallow water level (10mbgl) has observed in the eastern part of Pratapgarh block.

Depth to water level (m bgl)	Block wise area coverage (sq km) *					Total Area (sq km)
	Arnod	Chhoti Sadri	Dhariawad	Peepalkhoont	Pratapgarh	
< 10	24.4	-	-	-	4.7	29.1
10-20	389.4	502	38.7	0.7	518.7	1,449.5
> 20	133.5	60.3	405.4	428.4	417.9	1,445.5
Total	547.3	562.3	444.1	429.1	941.3	2,924.1

* The area covered in the derived maps is less than the total district area since the hills have been excluded from interpolation/contouring.



PLATE - VIII



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- State Boundary
- District Boundary
- Block Boundary
- Basin Boundary
- Roads:**
- National Highway
- State Highway
- Major District Road

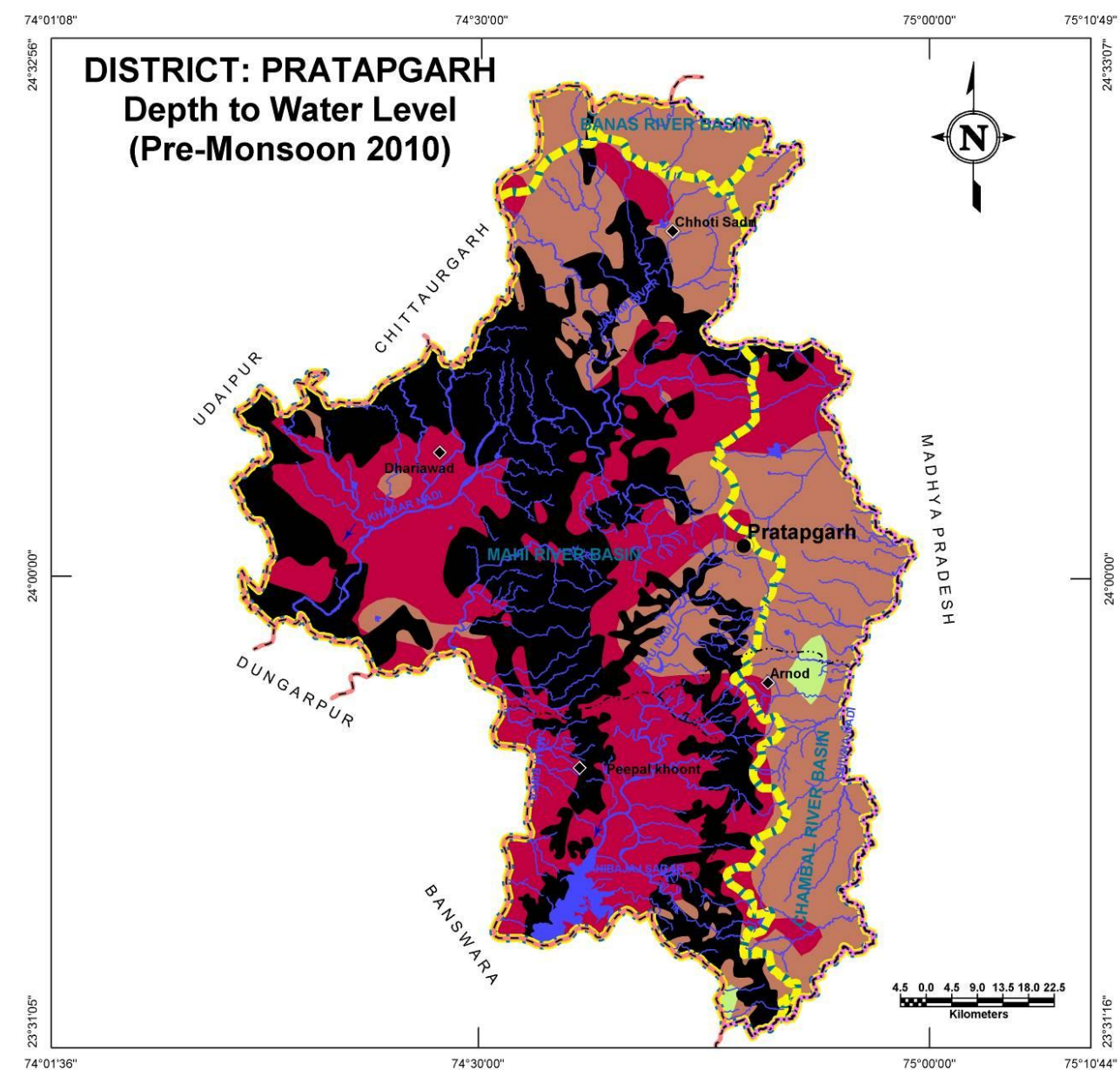
Water Bodies:

- River / Streams
- Ponds / Reservoirs
- Hills

Exploratory Wells:

- CGWB
- RGWD
- Ground Water Monitoring Stations:**
- CGWB
- RGWD

PLATE - IX



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- State Boundary
- District Boundary
- Block Boundary
- Basin Boundary
- Water Bodies:**
- River / Streams
- Ponds / Reservoirs
- Hills

Depth to Water Level (m bgl):

- < 10
- 10 - 20
- > 20

WATER TABLE ELEVATION (PRE MONSOON – 2010)

DISTRICT – PRATAPGARH

The flow directions inferred from water table contour map bring out that the drainage divide between Chambal basin and Mahi basin roughly coincides with regional ground water flow divide as well. The flow to the west of this drainage divide (i.e., within Mahi basin) is from east to west and in the Chambal basin part the flow of ground water is from west to east. In the northern part, the flow of ground water is in northerly direction, again relating well with Mahi-Banas drainage divide. Highest water table elevation reaching to >520m amsl is seen in central part of the district in Arnod and Pratapgarh blocks whereas, minimum elevation is very low (<180m amsl) in the Dhariawad block in western part of the district.

Table: Block wise area covered in each water table elevation range

Block Name	Block wise area coverage (sq km) within water table elevation range (m amsl)																Total Area (sq km)
	<180	180-200	200-220	220-240	240-260	260-280	280-300	300-320	320-340	340-360	360-380	380-400	400-440	440-480	480-520	>520	
Arnod	-	-	-	-	-	-	-	-	-	-	3.0	10.1	29.1	121.7	324.7	58.7	547.3
Chhoti Sadri	-	-	-	-	-	-	-	-	-	-	-	3.6	61.6	423.0	74.1	-	562.3
Dhariawad	-	47.3	172.5	155.5	49.0	15.0	4.3	0.5	-	-	-	-	-	-	-	-	444.1
Peepalkhoont	-	-	3.2	9.0	73.2	50.4	51.8	52.2	44.7	57.9	31.8	20.0	28.9	6.0	-	-	429.1
Pratapgarh	-	-	-	-	12.0	5.6	1.3	5.0	5.2	12.8	34.9	56.7	115.0	367.7	325.1	-	941.3
Total	-	47.3	175.7	164.5	134.2	71.0	57.4	57.7	49.9	70.7	69.7	90.4	234.6	918.4	723.9	58.7	2,924.1

WATER LEVEL FLUCTUATION (PRE TO POST MONSOON 2010)

A 2m contour interval adopted to visualize the ground water level fluctuation reveals a localized fall of more than 2 m in Chhoti Sadri area to rise of more than 8m also in Chhoti Sadri area itself as seen in Plate – XI. The –ve fluctuation areas (indicated by pink and red regions) are the areas where overexploitation is taking place, however these are small isolated pockets only. Most part of the district has shown a general to significant rise in ground water level in the post monsoon season with respect to pre monsoon season as general fluctuation range varies from 0m to 6m.

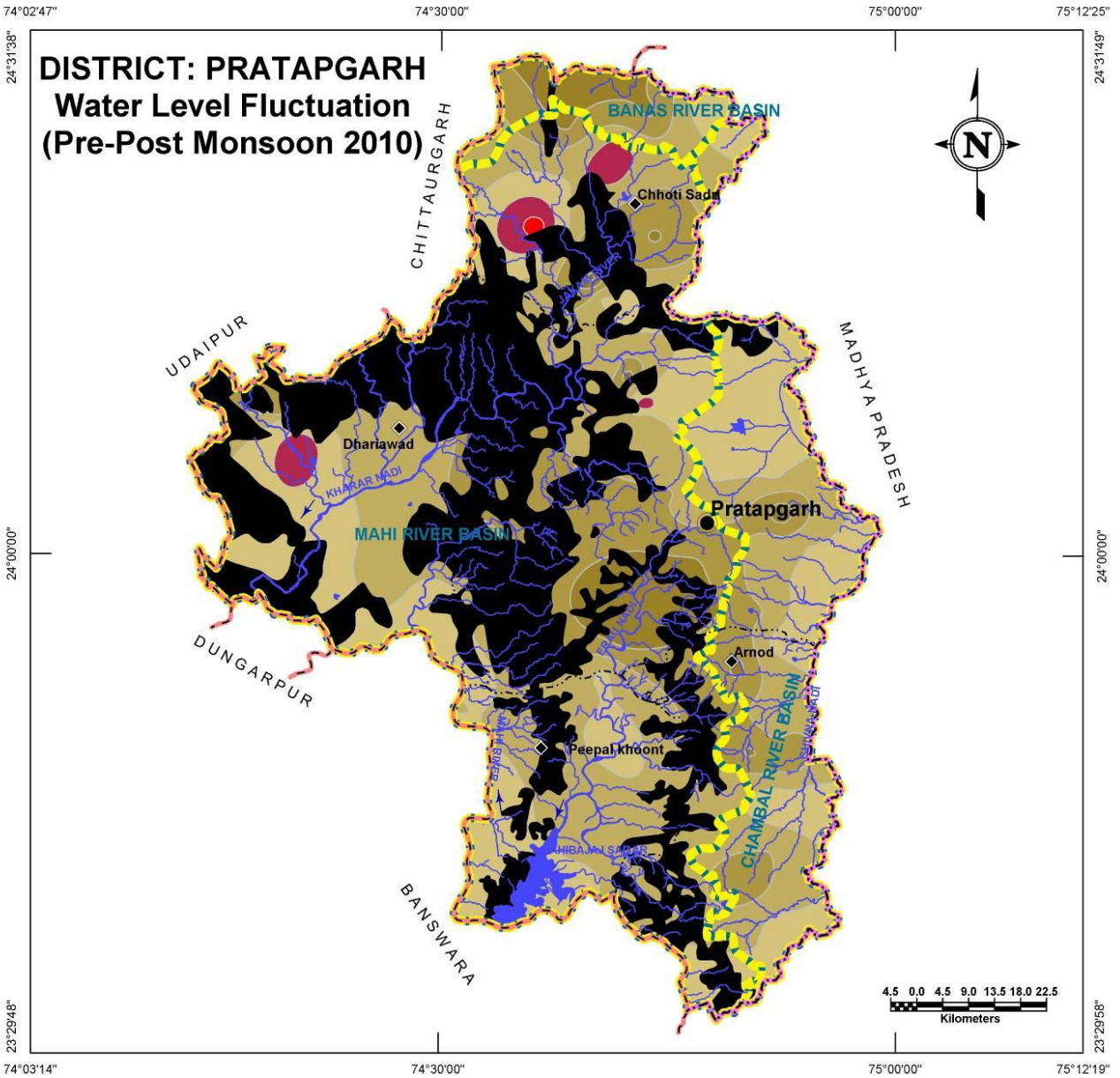
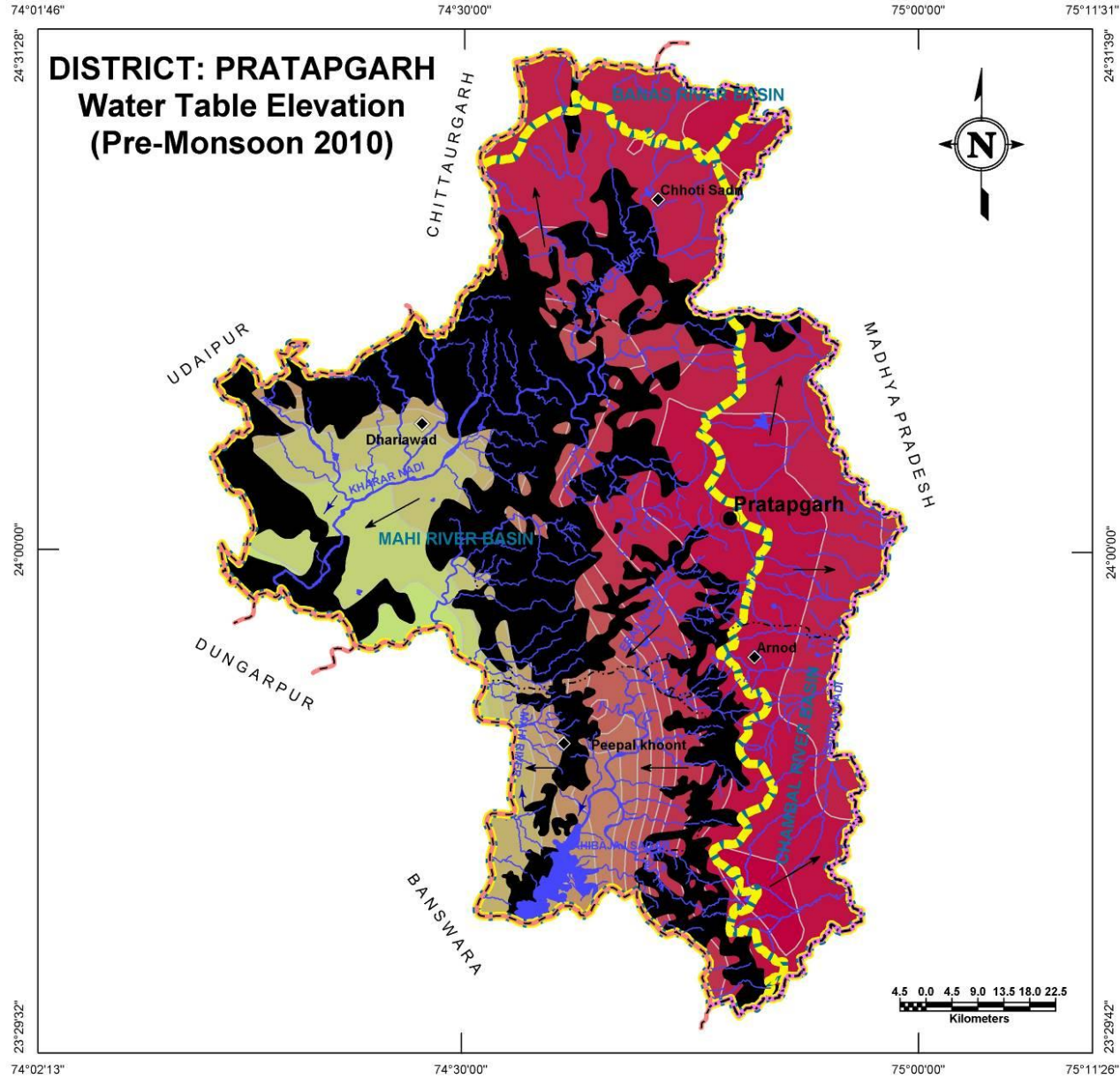
Table: Block wise area covered in each water fluctuation zone

Water level fluctuation range (m)	Block wise area coverage (sq km)					Total Area (sq km)
	Arnod	Chhoti Sadri	Dhariawad	Peepalkhoont	Pratapgarh	
< -2	-	3.4	-	-	-	3.4
-2 to 0	-	42.9	22.7	-	1.9	67.5
0 to 2	128.0	98.9	209.0	188.2	313.6	937.7
2 to 4	261.6	209.5	192.5	240.8	291.4	1,195.8
4 to 6	154.6	141.9	19.9	0.1	243.3	559.8
6 to 8	3.1	65.3	-	-	91.1	159.5
> 8	-	0.4	-	-	-	0.4
Total	547.3	562.3	444.1	429.1	941.3	2,924.1



PLATE - X

PLATE - XI



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- State Boundary
- District Boundary
- Block Boundary
- Basin Boundary**
- Water Bodies:**
- River / Streams
- Ponds / Reservoirs
- Hills
- Water Table Elevation**

Water Table Elevation (m amsl):

- < 180
- 180 - 200
- 200 - 220
- 220 - 240
- 240 - 260
- 260 - 280
- 280 - 300
- 300 - 320
- 320 - 340
- 340 - 360
- 360 - 380
- 380 - 400
- 400 - 440
- 440 - 480
- 480 - 520
- > 520

LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- State Boundary
- District Boundary
- Block Boundary
- Basin Boundary**
- Water Bodies:**
- River / Streams
- Ponds / Reservoirs
- Hills

Water Level Fluctuation (m):

- < -2
- 2 to 0
- 0 to 2
- 2 to 4
- 4 to 6
- 6 to 8
- > 8

GROUND WATER ELECTRICAL CONDUCTIVITY DISTRIBUTION

DISTRICT – PRATAPGARH

The Electrical conductivity (at 25°C) distribution map is presented in Plate – XII. The areas with low EC values in ground water (<2000 $\mu\text{S}/\text{cm}$) are shown in yellow color and occupies almost 99% of the district area indicating that, by and large the ground water in this region is suitable for domestic purpose. The areas with moderately high EC values (2000 - 4000 $\mu\text{S}/\text{cm}$) are shown in green color and occupy just 1% of the district area, occurring as small isolated patches in southern part of district.

Table: Block wise area of Electrical conductivity distribution

Electrical Conductivity Ranges ($\mu\text{S}/\text{cm}$ at 25°C) (Ave. of years 2005-09)	Block wise area coverage (sq km)										Total Area (sq km)
	Arnod		Chhoti Sadri		Dhariawad		Peepalkhoont		Pratapgarh		
	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	
< 2000	525.3	96.0	562.3	100.0	444.1	100.0	429.1	100.0	941.3	100.0	2,902.1
2000-4000	22.0	4.0	-	-	-	-	-	-	-	-	26.0
>4000	-	-	-	-	-	-	-	-	-	-	-
Total	547.3	100.0	562.3	100.0	444.1	100.0	429.1	100.0	941.3	100.0	2,924.1

GROUND WATER CHLORIDE DISTRIBUTION

The yellow colored regions in Plate – XIII are such areas where chloride concentration is low (<250 mg/l) that occupy approximately 83% of the district area and the ground water in this region is suitable for domestic purpose. The areas with moderately high chloride concentration (250-1000mg/l) are shown in green color and occupy approximately 17% of the district area, largely in the eastern part of Pratapgarh district. No area has shown presence of high chloride concentration in ground water.

Table: Block wise area of Chloride distribution

Chloride Concentration Range (mg/l) (Ave. of years 2005-09)	Block wise area coverage (sq km)										Total Area (sq km)
	Arnod		Chhoti Sadri		Dhariawad		Peepalkhoont		Pratapgarh		
	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	
< 250	283.0	52.0	560.5	100.0	444.1	100.0	428.3	100.0	719.1	76.0	2,435.0
250-1000	264.3	48.0	1.8	-	-	-	0.8	-	222.2	24.0	489.1
> 1000	-	-	-	-	-	-	-	-	-	-	-
Total	547.3	100.0	562.3	100.0	444.1	100.0	429.1	100.0	941.3	100.0	2,924.1

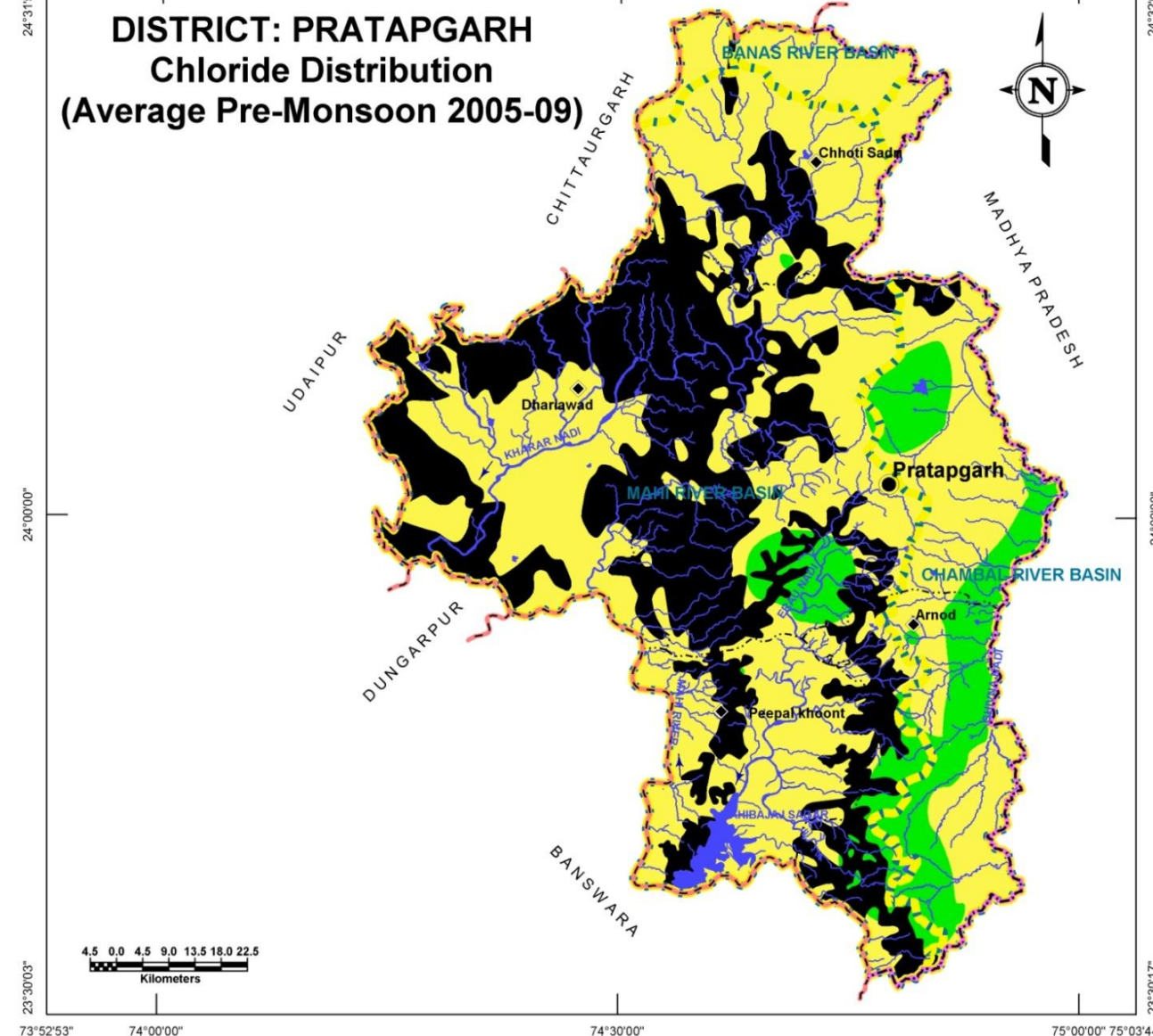
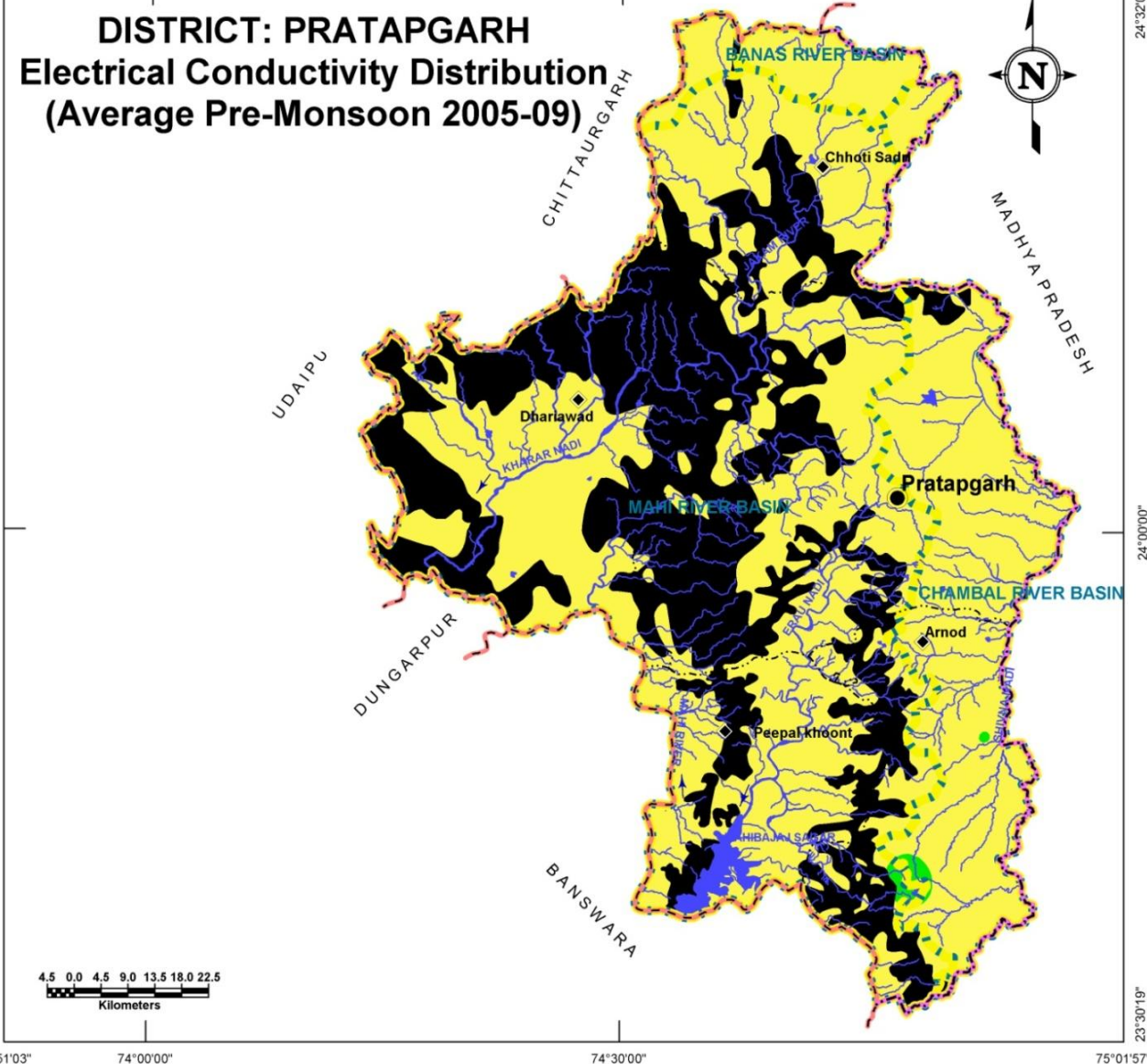


PLATE - XII

PLATE - XIII

DISTRICT: PRATAPGARH
Electrical Conductivity Distribution
(Average Pre-Monsoon 2005-09)

DISTRICT: PRATAPGARH
Chloride Distribution
(Average Pre-Monsoon 2005-09)



LEGEND

Admin Boundary:		Electrical Conductivity ($\mu\text{S}/\text{cm}$ at 25°C):	
District Headquarter			< 2000
Block Headquarter			2000-4000
State Boundary			
District Boundary			
Block Boundary			
Basin Boundary			
Water Bodies:			
River / Streams			
Ponds / Reservoirs			
Hills			

LEGEND

Admin Boundary:		Chloride Concentration (mg/l):	
District Headquarter			< 250
Block Headquarter			250 - 1000
State Boundary			
District Boundary			
Block Boundary			
Basin Boundary			
Water Bodies:			
River / Streams			
Ponds / Reservoirs			
Hills			

GROUND WATER FLUORIDE DISTRIBUTION

DISTRICT – PRATAPGARH

The Fluoride concentration map is presented in Plate – XIV. The areas with low concentration (i.e., >1.5 mg/l) are shown in yellow color and such areas are seen to occupy almost 99% of the district area which has ground water suitable for domestic purpose. The areas with moderately high concentration (1.5-3.0 mg/l) in green color patches and high Fluoride concentration (>3.0 mg/l) in red color patches together occupy less than 1% of the district area as small isolated pockets adjacent to hills in central and northern parts of the district.

Table: Block wise area of Fluoride distribution

Fluoride concentration Range(mg/l) (Ave. of years 2005-09)	Block wise area coverage (sq km)										Total Area (sq km)
	Arnod		Chhoti Sadri		Dhariawad		Peepalkhoont		Pratapgarh		
	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	
< 1.5	547.3	100.0	559.1	99.4	444.1	100.0	413.4	96.3	938.4	99.7	2,902.3
1.5-3.0	-	-	3.2	0.6	-	-	15.4	3.6	2.9	0.3	21.5
> 3.0	-	-	-	-	-	-	0.3	0.1	-	-	0.3
Total	547.3	100.0	562.3	100.0	444.1	100.0	429.1	100.0	941.3	100.0	2,924.1

GROUND WATER NITRATE DISTRIBUTION

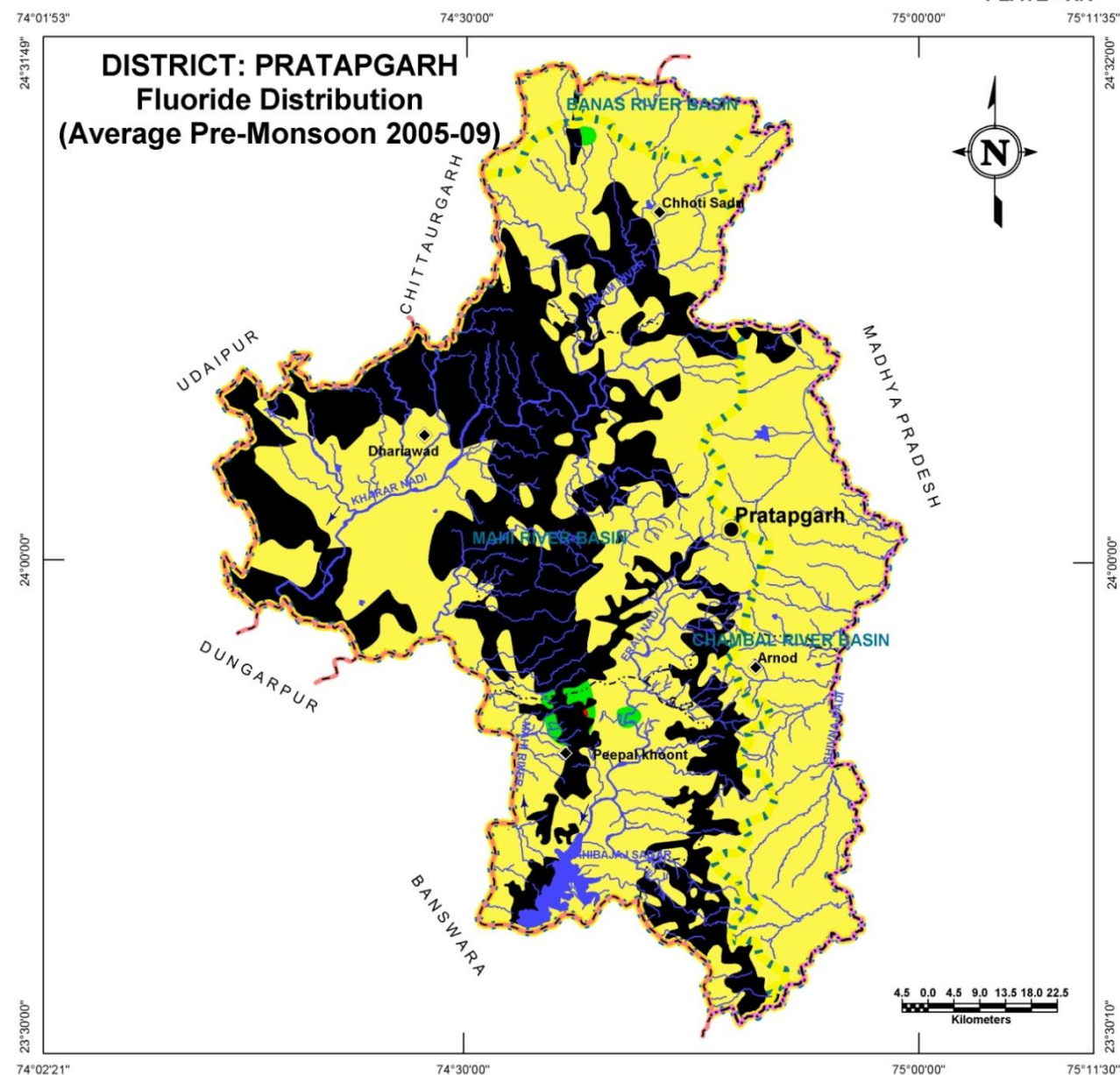
High nitrate concentration in ground water renders it unsuitable for agriculture purposes. Plate – XV shows distribution of Nitrate in ground water. Low nitrate concentration (<50 mg/l) is shown in yellow color and occupies approximately 48% of the district area which is suitable for agriculture purpose. The areas with moderately high nitrate concentration (50-100 mg/l) are shown in green color and occupy approximately 46% of the district area, largely northern part of the district. Remaining part of the district area is covered with high nitrate concentration (>100 mg/l) which is shown in red colored patches, which is not suitable for agriculture purpose however, such patches are either within hilly tract or adjacent to hills which may not be agricultural/crop field.

Table: Block wise area of Nitrate distribution

Nitrate concentration Range(mg/l) (Ave. of years 2005-09)	Block wise area coverage (sq km)										Total Area (sq km)
	Arnod		Chhoti Sadri		Dhariawad		Peepalkhoont		Pratapgarh		
	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	
< 50	238.8	43.6	180.8	32.2	205.2	46.4	391.2	91.1	396.0	42.0	1,412.0
50-100	297.4	54.4	280.0	49.8	238.9	53.6	36.3	8.5	488.2	51.9	1,340.8
>100	11.1	2.0	101.5	18.0	-	-	1.6	0.4	57.1	6.1	171.3
Total	547.3	100.0	562.3	100.0	444.1	100.0	429.1	100.0	941.3	100.0	2,924.1



PLATE - XIV



LEGEND

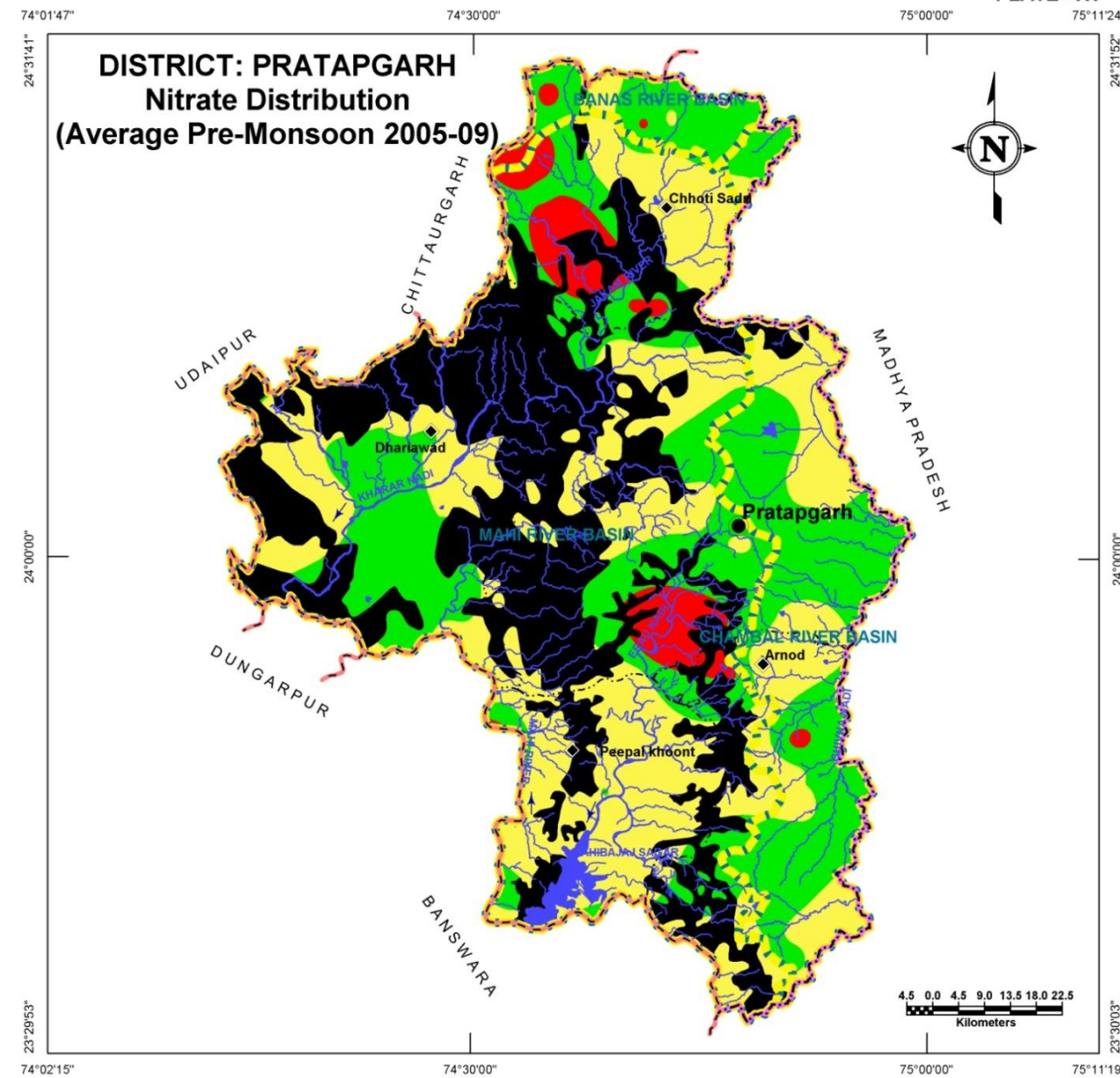
Admin Boundary:

- District Headquarter
- Block Headquarter
- State Boundary
- District Boundary
- Block Boundary
- Basin Boundary
- Water Bodies:
- River / Streams
- Ponds / Reservoirs
- Hills

Fluoride Concentration (mg/l):

- < 1.5
- 1.5-3.0
- > 3.0

PLATE - XV



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- State Boundary
- District Boundary
- Block Boundary
- Basin Boundary
- Water Bodies:
- River / Streams
- Ponds / Reservoirs
- Hills

Nitrate Concentration (mg/l):

- < 50
- 50-100
- > 100

DEPTH TO BEDROCK

DISTRICT – PRATAPGARH

From hydrogeological perspective, the beginning of massive bedrock has been considered for defining top of bedrock surface. There are either no alluvial covers over the bedrock or are very thin to be important from hydrogeological view point. Depth to bedrock map of Pratapgarh district (Plate – XVI) thus reveals wide variation of more than 20m below ground level reaching in areas adjacent to hills, to a maximum depth of more than 80m in the western part of the district. The major rocks types occurring in the district are Phyllites, Shale, Basalt and BGC. The bedrock occurs at shallow depths (less than 20m bgl) in and around Arnod, Chhoti Sadri and Pratapgarh blocks i.e., almost entire eastern part of the district. As we move to western periphery of the district, the depth of bedrock increases to a maximum of about 80m bgl. The depth of bedrock found more than 80 m bgl in Peepalkhoont block.

Depth to bedrock (m bgl)	Block wise area coverage (sq km)										Total Area (sq km)
	Arnod		Chhoti Sadri		Dhariawad		Peepalkhoont		Pratapgarh		
	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	
< 20	29.3	5.3	122.3	22.0	-	-	-	-	131.7	14.0	283.3
20-40	477.7	87.4	389.1	69.0	-	-	55.2	12.8	729.4	77.5	1,651.4
40-60	38.7	7.0	50.9	9.0	331.1	74.5	302.2	70.4	76.1	8.5	799.0
60-80	1.6	0.3	-	-	113.0	25.5	64.1	15.0	4.1	-	182.8
> 80	-	-	-	-	-	-	7.6	1.8	-	-	7.6
Total	547.3	100.0	562.3	100.0	444.1	100.0	429.1	100.0	941.3	100.0	2,924.1

UNCONFINED AQUIFER

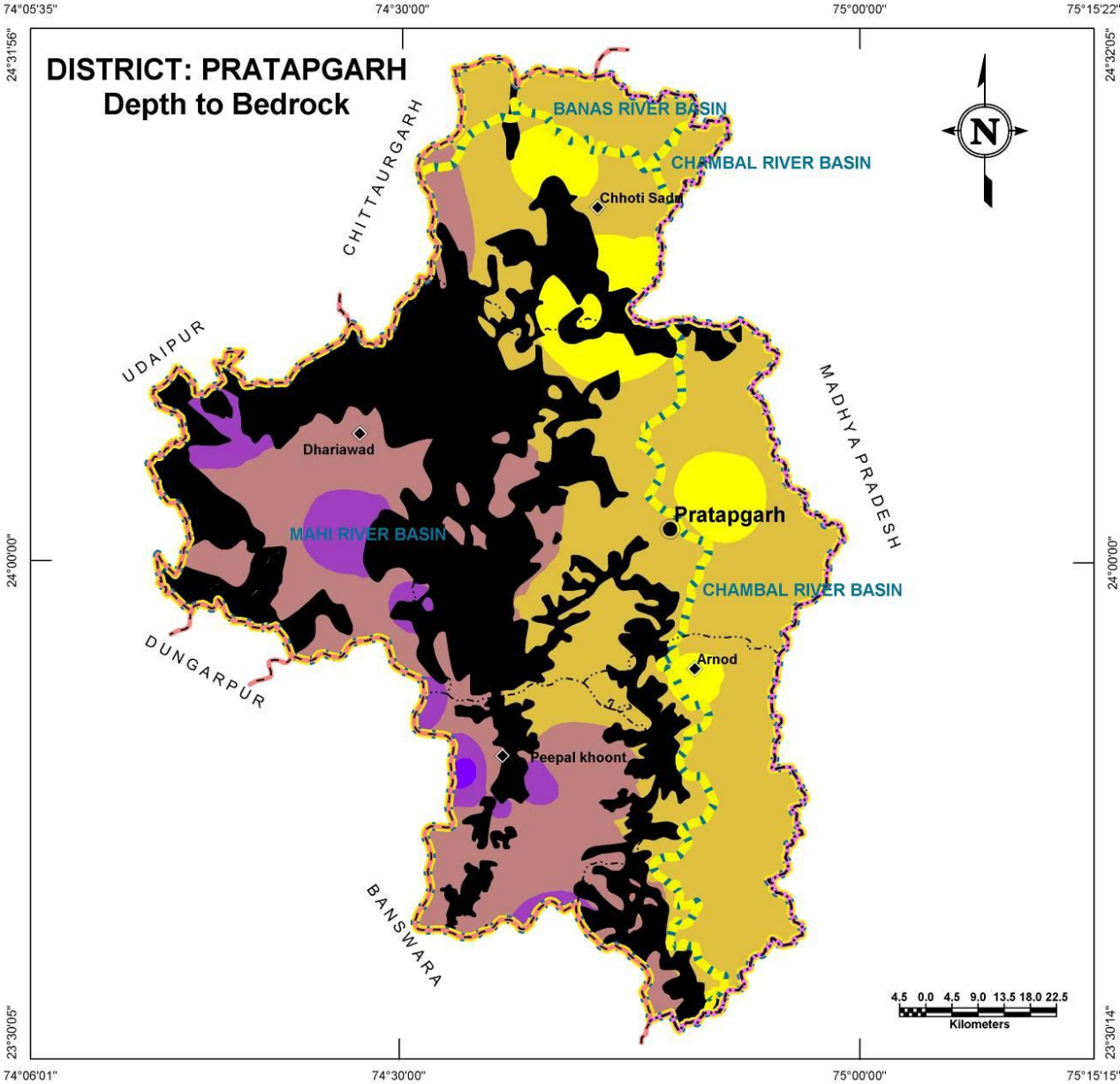
Unconfined aquifer hardrock areas

Aquifers in the district are predominantly formed in weathered, fractured and jointed rock formations occurring at shallower depths and these constitute good unconfined aquifers. The thickness of aquifers thus formed varies from less than 10m to around 50m. Isolated parts have very thick zones especially in Peepalkhoont and Pratapgarh where the thickness of aquifer in unconfined condition is seen to be more than 40m. Rest of the blocks have moderate to low thickness of aquifers in hardrock reaching to a maximum of 20m to 30m.

Unconfined aquifer Thickness (m)	Block wise Area coverage (sq km)					Total Area (sq km)
	Arnod	Chhoti Sadri	Dhariawad	Peepalkhoont	Pratapgarh	
< 10	381.9	496.2	0.4	211.5	529.6	1,619.6
10-20	165.1	66.1	93.2	184.2	196.6	705.2
20-30	-	-	289.1	32.1	174.2	495.4
30-40	0.3	-	61.4	1.2	38.9	101.8
> 40	-	-	-	0.1	2.0	2.1
Total	547.3	562.3	444.1	429.1	941.3	2,924.1



PLATE - XVI



LEGEND

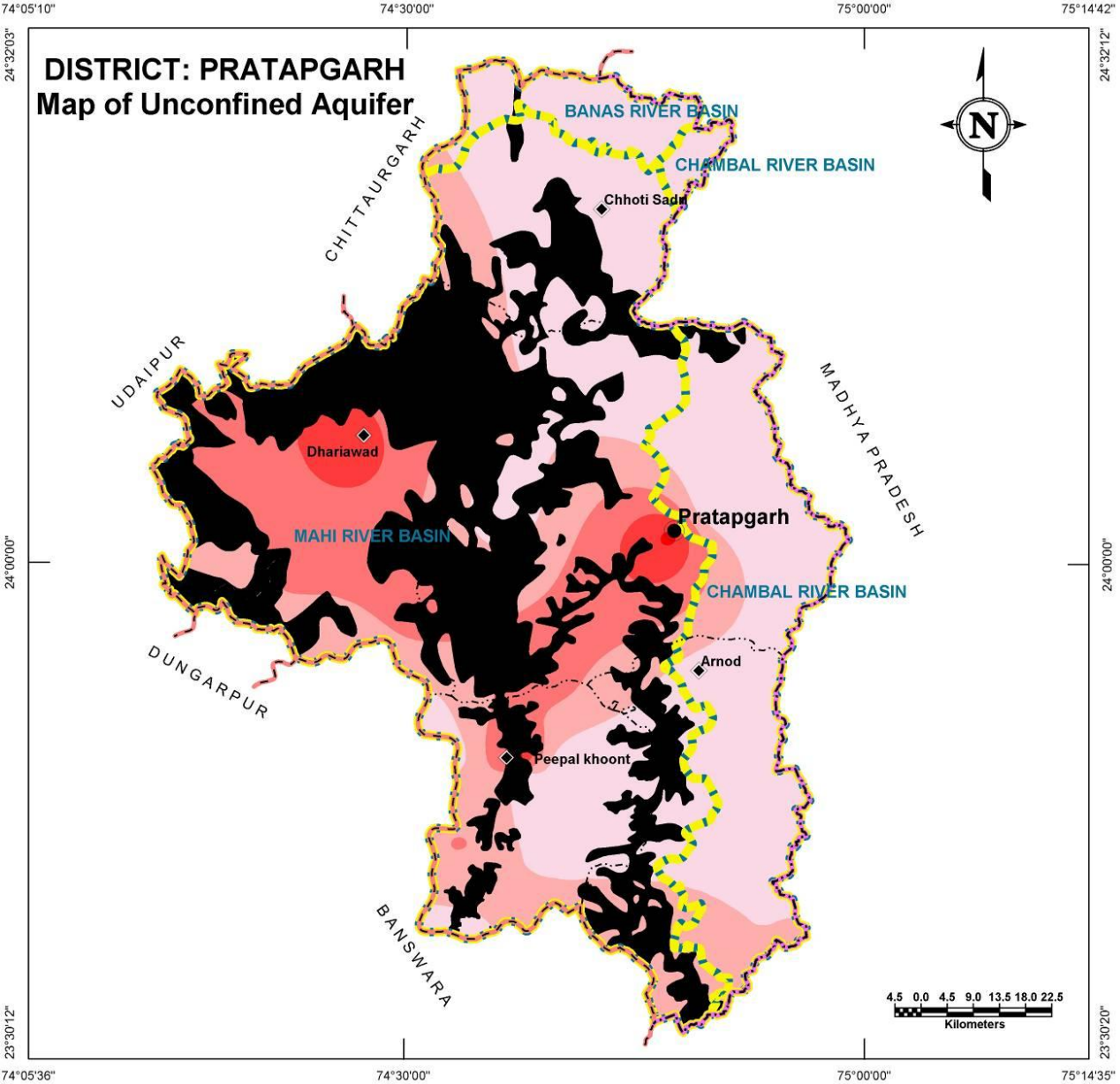
Admin Boundary:

- District Headquarter
- Block Headquarter
- State Boundary
- District Boundary
- Block Boundary
- Basin Boundary
- Hills

Depth to Bedrock(m bgl):

- <20
- 20-40
- 40-60
- 60-80
- >80

PLATE - XVII



Admin Boundary:

- District Headquarter
- Block Headquarter
- State Boundary
- District Boundary
- Block Boundary
- Basin Boundary
- Hills

Aquifer Thickness(m):

- Weathered & Fractured Zone in Hardrock Area**
- <10
 - 10-20
 - 20-30
 - 30-40
 - >40

Glossary of terms

S. No.	Technical Terms	Definition
1	AQUIFER	A saturated geological formation which has good permeability to supply sufficient quantity of water to a Tube well, well or spring.
2	ARID CLIMATE	Climate characterized by high evaporation and low precipitation.
3	ARTIFICIAL RECHARGE	Addition of water to a groundwater reservoir by man-made activity
4	CLIMATE	The sum total of all atmospheric or meteorological influences principally temperature, moisture, wind, pressure and evaporation of a region.
5	CONFINED AQUIFER	A water bearing strata having confined impermeable overburden. In this aquifer, water level represents the piezometric head.
6	CONTAMINATION	Introduction of undesirable substance, normally not found in water, which renders the water unfit for its intended use.
7	DRAWDOWN	The drawdown is the depth by which water level is lowered.
8	FRESH WATER	Water suitable for drinking purpose.
9	GROUND WATER	Water found below the land surface.
10	GROUND WATER BASIN	A hydro-geologic unit containing one large aquifer or several connected and interrelated aquifers.
11	GROUNDWATER RECHARGE	The natural infiltration of surface water into the ground.
12	HARD WATER	The water which does not produce sufficient foam with soap.
13	HYDRAULIC CONDUCTIVITY	A constant that serves as a measure of permeability of porous medium.
14	HYDROGEOLOGY	The science related with the ground water.
15	HUMID CLIMATE	The area having high moisture content.
16	ISOHYET	A line of equal amount of rainfall.
17	METEOROLOGY	Science of the atmosphere.
18	PERCOLATION	It is flow through a porous substance.
19	PERMEABILITY	The property or capacity of a soil or rock for transmitting water.
20	pH	Value of hydrogen-ion concentration in water. Used as an indicator of acidity (pH < 7) or alkalinity (pH > 7).
21	PIEZOMETRIC HEAD	Elevation to which water will rise in a piezometers.
22	RECHARGE	It is a natural or artificial process by which water is added from outside to the aquifer.
23	SAFE YIELD	Amount of water which can be extracted from groundwater without producing undesirable effect.
24	SALINITY	Concentration of dissolved salts.
25	SEMI-ARID	An area is considered semiarid having annual rainfall between 10-20 inches.
26	SEMI-CONFINED AQUIFER	Aquifer overlain and/or underlain by a relatively thin semi-pervious layer.
27	SPECIFIC YIELD	Quantity of water which is released by a formation after its complete saturation.
28	TOTAL DISSOLVED SOLIDS	Total weight of dissolved mineral constituents in water per unit volume (or weight) of water in the sample.

(Contd...)

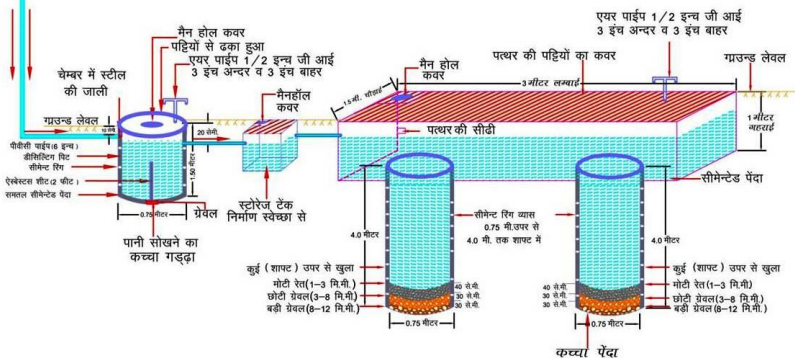
S. No.	Technical Terms	Definition
29	TRANSMISSIBILITY	It is defined as the rate of flow through an aquifer of unit width and total saturation depth under unit hydraulic gradient. It is equal to product of full saturation depth of aquifer and its coefficient of permeability.
30	UNCONFINED AQUIFER	A water bearing formation having permeable overburden. The water table forms the upper boundary of the aquifer.
31	UNSATURATED ZONE	The zone below the land surface in which pore space contains both water and air.
32	WATER CONSERVATION	Optimal use and proper storage of water.
33	WATER RESOURCES	Availability of surface and ground water.
34	WATER RESOURCES MANAGEMENT	Planned development, distribution and use of water resources.
35	WATER TABLE	Water table is the upper surface of the zone of saturation at atmospheric pressure.
36	ZONE OF SATURATION	The ground in which all pores are completely filled with water.
37	ELECTRICAL CONDUCTIVITY	Flow of free ions in the water at 25C mu/cm.
38	CROSS SECTION	A Vertical Projection showing sub-surface formations encountered in a specific plane.
39	3-D PICTURE	A structure showing all three dimensions i.e. length, width and depth.
40	GWD	Ground Water Department
41	CGWB	Central Ground Water Board
42	CGWA	Central Ground Water Authority
43	SWRPD	State Water Resources Planning Department
44	EU-SPP	European Union State Partnership Programme
45	TOPOGRAPHY	Details of drainage lines and physical features of land surface on a map.
46	GEOLOGY	The science related with the Earth.
47	GEOMORPHOLOGY	The description and interpretation of land forms.
48	PRE MONSOON SURVEY	Monitoring of Ground Water level from the selected DKW/Piezometer before Monsoon (carried out between 15th May to 15th June)
49	POST-MONSOON SURVEY	Monitoring of Ground Water level from the selected DKW/Piezometer after Monsoon (carried out between 15th October to 15th November)
50	PIEZOMETER	A non-pumping small diameter bore hole used for monitoring of static water level.
51	GROUND WATER FLUCTUATION	Change in static water level below ground level.
52	WATER TABLE	The static water level found in unconfined aquifer.
53	DEPTH OF BED ROCK	Hard & compact rock encountered below land Surface.
54	G.W. MONITORING STATION	Dug wells selected on grid basis for monitoring of state water level.
55	EOLIAN DEPOSITS	Wind-blown sand deposits



भवन छत क्षेत्रफल 300 से 500 वर्गमीटर तक
निर्माण किये जाने वाले मुख्य भाग एवं डिज़ाईन

चित्र-4

- PVC पाईप 6" व्यास
- सीमेंट रिंग से निर्मित डीसिल्टिंग पिट (0.75 मी व्यास x 1.50 मी गहरा)
- रीचार्ज टैंक 1.5 मी चौड़ा x 3 मी लम्बा x 1 मी गहरा
- सीमेंट रिंग से निर्मित शाफ्ट (0.75 मी व्यास x 4 मी गहरा) (संख्या 2)
- संरचना की अनुमानित लागत रु 24,000 अधिक
- वार्षिक पुनर्भरित जल लगभग 2,00,000 लीटर
- 20 वर्षों में पुनर्भरित जल लगभग 40,00,000 लीटर
- पुनर्भरित जल की लागत 1 पैसे प्रति लीटर से कम



भूजल में घुले मुख्य तत्वों की अधिकता का मानव शरीर पर दुष्प्रभाव

बोरॉन-स्नायु तन्त्र पर प्रभाव

फ्लोराइड - दंत क्षरण

क्लोराइड-सोडियम के साथ मिलकर उच्च रक्त चाप

सोडियम-हृदय, गुर्दा व रक्त परिसंचरण रोगों से ग्रसित लोगों को हानिकारक

कैल्शियम-जोड़ों में कड़ापन

नाइट्रेट-नवजात शिशुओं में ब्लू बेबी बीमारी (मेथेमोग्लोबिनिमिया)

आर्सेनिक-त्वचा रोग, कैंसर

सल्फेट-अधिकता में मैग्नेशियम के साथ मिलकर दस्तावर

लेड-बच्चों के शारीरिक व मानसिक विकास में बाधा वयस्कों में गुर्दे के रोग

आयरन-आयरन जीवाणु से आमाशय संबंधी रोग

फ्लोराइड-जोड़ों में अकड़न, हड्डियों में मुड़ाव



केन्द्रीय भूमि जल बोर्ड,
पश्चिमी क्षेत्र, जयपुर
जल संसाधन मंत्रालय
भारत सरकार
e-mail: cgwbwr@sancharnet.in



भूजल अमूल्य है इसे प्रदूषित न करें।

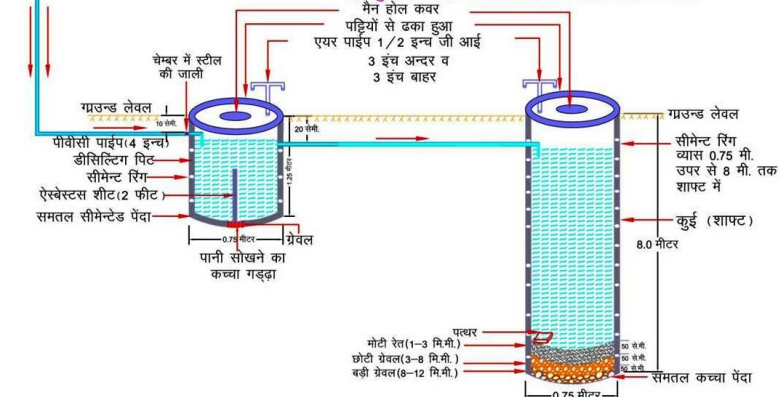


भवन छत क्षेत्रफल 100 से 200 वर्गमीटर तक

चित्र-2

निर्माण किये जाने वाले मुख्य भाग एवं डिज़ाईन

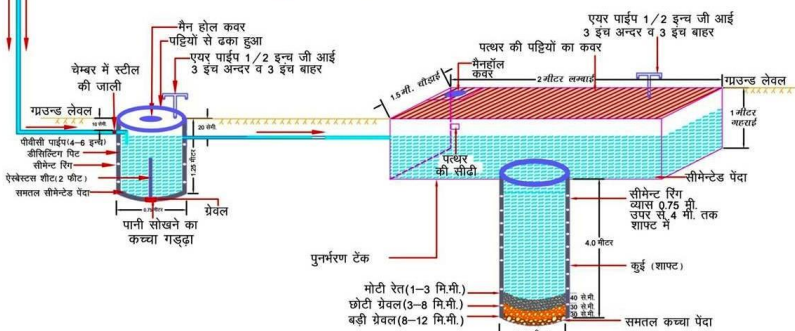
- PVC पाईप 4" व्यास
- सीमेंट रिंग से निर्मित डीसिल्टिंग पिट (0.75 मी व्यास x 1.25 मी गहरा)
- सीमेंट रिंग से निर्मित शाफ्ट (0.75 मी व्यास x 4 मी गहरा)
- संरचना की अनुमानित लागत रु 11,000-12,000
- वार्षिक पुनर्भरित जल लगभग 83,000 लीटर
- 20 वर्षों में पुनर्भरित जल लगभग 16,64,000 लीटर
- पुनर्भरित जल की लागत 1 पैसे प्रति लीटर से कम



भवन छत क्षेत्रफल 200 से 300 वर्गमीटर तक
निर्माण किये जाने वाले मुख्य भाग एवं डिज़ाईन

चित्र-3

- PVC पाईप 4" - 6" व्यास
- सीमेंट रिंग से निर्मित डीसिल्टिंग पिट (0.75 मी व्यास x 1.25 मी गहरा)
- रीचार्ज टैंक 1.5 मी चौड़ा x 2 मी लम्बा x 1 मी गहरा
- सीमेंट रिंग से निर्मित शाफ्ट (0.75 मी व्यास x 4 मी गहरा)
- संरचना की अनुमानित लागत रु 15,000-16,000
- वार्षिक पुनर्भरित जल लगभग 1,25,000 लीटर
- 20 वर्षों में पुनर्भरित जल लगभग 25,00,000 लीटर
- पुनर्भरित जल की लागत 1 पैसे प्रति लीटर से कम

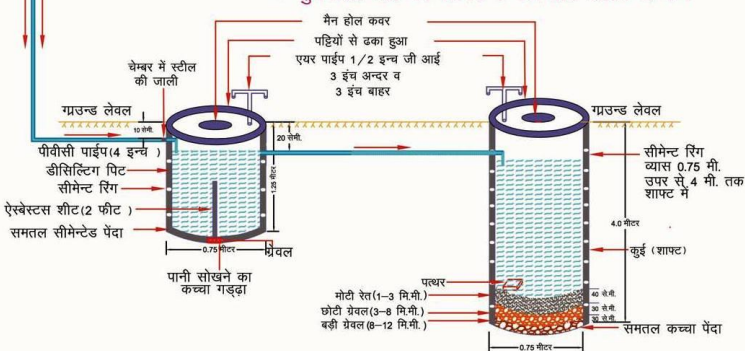


भवन छत क्षेत्रफल 100 वर्गमीटर तक

चित्र-1

निर्माण किये जाने वाले मुख्य भाग एवं डिज़ाईन

- PVC पाईप 4" व्यास
- सीमेंट रिंग से निर्मित डीसिल्टिंग पिट (0.75 मी व्यास x 1.25 मी गहरा)
- सीमेंट रिंग से निर्मित शाफ्ट (0.75 मी व्यास x 4 मी गहरा)
- संरचना की अनुमानित लागत रु 7000-8000
- वार्षिक पुनर्भरित जल लगभग 40,000 लीटर
- 20 वर्षों में पुनर्भरित जल लगभग 8,00,000 लीटर
- पुनर्भरित जल की लागत 1 पैसे प्रति लीटर से कम



Myths and Facts about Ground Water

S No	Myths	Facts
1	What is Ground Water • an underground lake • a net work of underground rivers • a bowl filled with water	Water which occurs below the land in geological formations/rocks is Ground water
2	Ground Water occurs everywhere beneath the Land Surface	Not really, it depends on the nature of rock formation
3	There is a relationship between ground water and surface water	Not all the places. Near streams/rivers there is relation
4	Groundwater is not renewable resource	It is renewable source and every year it is being recharged through rain/applied irrigation etc
5	Ground water is unlimited and deeper you drill more discharge	It is limited to annual recharge from rain/applied irrigation. The discharge may not increase if you go deeper
6	Ground Water moves rapidly	The movement of ground water is very slow
7	Ground water pumped from wells is thousands of years old	Generally the ground water being tapped through wells is a few years old
8	If water taste good—it is safe to drink	It may have other chemicals e.g. fluoride, nitrates etc which are harmful
9	Water from free flowing tube wells is very pure	This water can also be contaminated so test before use
10	If I recharge my TW/DW/HP it will not benefit me	It will also benefit you and also adjoining wells
11	There is no static ground water resources in Rajasthan	Rajasthan is also having Static GW resources, and being tapped in most of areas as GW annual withdrawal is more than annual recharge
12	I cannot meet annual cooking and drinking water requirement by rain water harvesting	The water requirement for drinking and cooking is only 8 lit/day. You can harvest this water for family of 5 persons from roof top or paved area of 75 Sq m to meet annual requirement
13	You can increase ground water recharge	This can be done by harvesting the rain water and storing in sub surface reservoir (GW) by constructing the recharge structures
14	You cannot use abandoned TW/HP/DW for ground water recharge	These should be used as recharge structures as harvested rain water is directly put into GW reservoir
15	Putting waste near HP/TW will not cause any problem	Such actions will pollute wells and water



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